

Multilayer Formation, Interfacial Binding, and Stability of Self-Assembled Monolayers in Perovskite Solar Cells

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Materials

All materials were used as received without further purification. MeO-2PACz, 2PACz, 4PADCB, lead iodide (PbI_2 , 99.99%), lead bromide (PbBr_2 , >98.0%), and BCP were purchased from TCI. Formamidinium iodide (FAI, >99.99%) was purchased from Great Cell Solar. Cesium iodide (CsI , 99.999%), lithium fluoride (LiF), anhydrous *N,N*-dimethylformamide (DMF, 99.8%), dimethyl sulfoxide (DMSO, 99.9%), ethanol (99.5%), and chlorobenzene (99.8%) were purchased from Sigma-Aldrich. C_{60} was purchased from Fisher Scientific. Patterned and unpatterned ITO substrates were purchased from All about Materials and Glass (AMG) and Ossila, respectively.

Methods

Density Functional Theory (DFT) calculations

All DFT calculations were completed using the Vienna Ab Initio Simulation Package (VASP) with the Perdew-Burke-Ernzerhof functional.¹⁻⁴ Grimme's D3 correction was used to account for long-range van der Waals interactions.⁵ A plane wave energy cutoff of 450 eV was used for valence electron expansion, while the core electrons were accounted for using projected augmented wave potentials. Spin polarized calculations were used for the deprotonated SAM molecules. Due to the large supercell size, only the gamma point was needed for *k*-point sampling for all calculations.

Gas-phase calculations involving molecules, including SAM molecules, ethanol, DMF, DMSO, a SAM-ethanol dimer, a SAM-DMF dimer, and a SAM-DMSO dimer were placed in vacuum separated by 40 Å in each direction to minimize periodic interactions. Each structure was relaxed by geometry optimization, with forces converged to <0.03 eV/Å. The ITO surface (simulated as the (111) facet of indium oxide because of the dilute concentrations of Sn) was directly taken from the work of Fei et al. and reoptimized in VASP⁶. The (111) facet was selected in this work as it is commonly used to model In_2O_3 surfaces, owing to the thermodynamic stability of the (222) facet⁷⁻¹¹, which can be equivalently represented by the (111) facet for cubic cells^{6,9,12-15}. Periodic images in the *z*-direction were separated by a vacuum layer of 30 Å to minimize interactions.

For the adsorption studies, all SAM adsorbates and the two topmost layers of ITO were relaxed by geometry optimization, with forces converged to <0.05 eV/Å. The hydroxyl-terminated slabs were relaxed to a force tolerance of <0.15 eV/Å. We found that it was considerably more difficult to

optimize these structures below a force tolerance of 0.10 eV/Å. However, within a range of 0.10 eV/Å to 0.15 eV/Å, the energy of the system was changing by only 0.007 eV, indicating the shallow nature of the potential energy surface. All energies were computed at 0 K without thermochemistry or solvation corrections. This means that all reaction energies currently neglect entropic losses when a molecule adsorbs and entropic gains in the removal process, as well as the impact of surrounding solvent on the adsorption/removal process. Therefore, the calculations tend to overestimate the binding strength of all species to the surface and underestimate the favorability of their removal. As such, we simply focused on the relative binding strengths and removal energies of the different solvents (i.e., whether one solvent yielded more favorable removal energy than another).

We considered two primary mechanisms when computing the adsorption energy of the SAM molecule to In₂O₃(111). The first mechanism involves reactive adsorption ($\Delta E_{reactive}$), where the SAM donates two hydrogens to hydroxyl groups on the surface and forms water (**Eq. 1**). In this equation, $E_{SAM^*,A}$ and E_{H_2O} are the total electronic energies of an adsorbed SAM (minus two hydrogens on the phosphate group) on In₂O₃(111) and a water molecule, respectively. E_{SAM} and $E_{In_2O_3-2OH}$ are the total electronic energies of an isolated SAM and In₂O₃(111) with two indium atoms terminated with hydroxyl groups. This methodology was adapted from Fei et al.⁶ The reactively adsorbed SAM molecules was modeled by orienting the molecule perpendicular to the surface such that the phosphonic acid group (without hydrogens) was directly interacting with the In atoms. The system was then optimized with DFT. Because the ITO surface may not be fully hydroxylated, it is plausible that a SAM molecule forms only a single covalent bond. Accordingly, a configuration in which the phosphonic acid forms one covalent bond and one hydrogen bond with its two hydroxyl groups was also modeled, and **Eq. 1 and 2** were modified accordingly to account for the one hydroxyl termination, the formation of a single water molecule, and a SAM molecule with one remaining hydroxyl group.

We also directly computed the binding energy, ($\Delta E_{b,reactive}$) of a deprotonated SAM molecule using **Eq. 2**, where $E_{SAM,A}$ and $E_{In_2O_3}$ are the energies of an isolated deprotonated SAM and In₂O₃(111) with no adsorbates, respectively. This binding energy yielded a better assessment of the strength of the bound SAM molecule following deprotonation. Nonetheless, the entire

heterocondensation reaction (i.e., including deprotonation and water formation) is thermodynamically favorable, with an overall energy of -2.74 eV, -2.27 eV, and -2.59 eV for MeO-2PACz, 2PACz, and 4PADCB, respectively.

$$\Delta E_{reactive} = (E_{SAM^*,A} + 2E_{H_2O}) - (E_{SAM} + E_{In_2O_3-2OH}) \text{ (Eq. 1)}$$

$$\Delta E_{b,reactive} = E_{SAM^*,A} - (E_{SAM,A} + E_{In_2O_3}) \text{ (Eq. 2)}$$

For non-reactive binding energy of the SAM molecule to $In_2O_3(111)$ ($\Delta E_{b,non-reactive}$), such as through hydrogen bonding or the carboxyl group, we only accounted for the binding energy of the SAM molecule to an $In_2O_3(111)$ without hydroxyl terminations. The non-reactive adsorption modes were modeled by placing the molecule perpendicular to the surface so that the phosphonic acid group could hydrogen bond with the surface or orienting it parallel to the surface so that the carbazole group could interact with the surface. Both modes were optimized with DFT to obtain their ground-state energies. As shown in **Eq. 3**, $E_{In_2O_3}$ and $E_{SAM^*,B}$ are the energies of the In_2O_3 surface without hydroxyl terminations and an adsorbed SAM molecule with its hydrogens, respectively.

$$\Delta E_{b,non-reactive} = E_{SAM^*,B} - (E_{SAM} + E_{In_2O_3}) \text{ (Eq. 3)}$$

We computed the energy to remove a SAM molecule with either ethanol, DMF, or DMSO through **Eq. 4**, where E_{SAM-M} and E_M are the energies of an isolated SAM-M dimer and an isolated solvent molecule (M = ethanol, DMF, or DMSO), respectively. The different schematics of the SAM-M dimers can be found in **Figs. S7–9**.

$$\Delta E_{removal} = (E_{SAM-M} + E_{In_2O_3}) - (E_M + E_{SAM^*}) \text{ (Eq. 4)}$$

SAM fabrication

ITO substrates were sequentially sonicated for 15 min each in aqueous detergent, deionized water, acetone, and isopropyl alcohol. After drying with nitrogen, the substrates underwent a 1-min

oxygen plasma treatment and were immediately transferred to a nitrogen-filled glovebox. 60 μL of SAM solution (1.5 mM in anhydrous ethanol) was spin-coated onto the ITO substrates at 4000 rpm for 30 s, followed by annealing at 100 $^{\circ}\text{C}$ for 10 min. For the first solvent washing step, the annealed SAM films were returned to the spin-coater and spun at 4000 rpm for 30 s. Approximately 5 s into spinning, after peak rpm was reached, 1 mL of rinsing solvent was dropped onto the film. The rinsed films were then annealed at 100 $^{\circ}\text{C}$ for 5 min. This washing process was repeated twice.

SAM Redeposition

For the redeposition strategy, SAM molecules were redeposited onto DMF-rinsed films using the same spin-coating and annealing procedure as initially completed (60 μL , 4000 rpm, 30 s, followed by annealing at 100 $^{\circ}\text{C}$ for 10 min). For the second rinsing process, two additional DMF solvent washes were performed using the same procedure as the initial wash (1 mL of solvent dispensed 5 s into spinning followed by annealing at 100 $^{\circ}\text{C}$ for 5 min).

Perovskite film fabrication

Perovskite precursor was prepared using the composition $\text{Cs}_{0.06}\text{FA}_{0.94}\text{PbI}_3$, with a concentration of 1.5 M. Specifically, CsI (19.5 mg), FAI (219.5 mg), and PbI_2 (656.9 mg) were dissolved in 1 mL of a DMF:DMSO (4:1 v/v) solvent mixture and vortexed for two hours to ensure complete dissolution. The precursor solution was then filtered through a 0.22 μm polytetrafluoroethylene membrane before use. For perovskite film fabrication, 60 μL of the precursor solution was deposited onto ITO substrates. The solution was then spin-coated at 5000 rpm for 35 s. With 10 s remaining in the spin cycle, 200 μL of chlorobenzene was dropped onto the substrate. The resulting films appeared a transparent yellow color. They were then annealed at 150 $^{\circ}\text{C}$ for 20 minutes, developing a black, shiny finish.

Device fabrication

Once the patterned ITO/SAM/Perovskite films were fabricated, they were then transferred to the thermal evaporator (Angstrom Engineering) without ambient exposure. LiF (1 nm), C_{60} (30 nm) and BCP (6 nm) were deposited sequentially with rates of 0.1 $\text{\AA s}^{-1}$, 0.25 $\text{\AA s}^{-1}$, and 0.2 $\text{\AA s}^{-1}$, respectively. Finally, 150 nm of Ag was deposited at a rate of 1.2 $\text{\AA s}^{-1}$ on top of BCP through a

shadow mask. For stability measurements, BCP was replaced with ALD-deposited SnO₂. Deposition of SnO₂ was carried out on the Veeco Fiji ALD at the Robert H. Lurie Nanofabrication Facility. H₂O and TDMASn were used as oxygen and tin precursors, respectively. Precursor and substrate temperatures were set to 68 and 130 °C, respectively, while the carrier gas was 110 sccm Ar. Pulse and purge times were 0.06 and 20 s for H₂O and 0.25 and 20 s for TDMASn. The total number of deposition cycles was 238, corresponding to 20 nm SnO₂. After Ag deposition, devices being tested for stability were encapsulated before the start of the measurement.

Device characterization

The current–voltage characteristics of the solar cells were measured using a Keithley 2450 source meter under illumination from a G2V Optics PicoTM solar simulator (AM1.5G, 1.0 Sun). The device active area was 0.06 cm².

X-ray Photoelectron Spectroscopy (XPS) measurements

XPS measurements were conducted on prepared substrates using a Kratos Axis Supra+ equipped with a monochromatic Al K α source at the Michigan Center for Materials Characterization. Data processing and analysis were performed using CASA XPS.

Cyclic voltammetry (CV) measurements

CV measurements were performed using a three-electrode cell with a Potentiostat (Squidstat Plus). A platinum plate was used as the counter electrode, and an Ag/AgNO₃ electrode was used as the reference electrode. SAM-covered ITO glasses served as the working electrodes. The measurements were conducted under a nitrogen atmosphere in an anhydrous and nitrogen-saturated o-DCB solution, with 0.1 M tetrabutylammonium hexafluorophosphate ($nBu_4N^+PF_6^-$) as the supporting electrolyte.

The effective coverage of the SAM on the ITO surface was determined using **Eq. 5**, which describes the linear dependence of the oxidative or reductive peak intensity on the scan rate¹⁶:

$$i_p = \frac{n^2 F^2}{4RTN_A} v A \Gamma \quad (\text{Eq. 5})$$

where i_p is the peak current (A), v is the voltage scan rate (mV s^{-1}), n is the number of electrons transferred (in this case, one), F is the Faraday constant ($96,485 \text{ C mol}^{-1}$), R is the universal gas constant ($8.3144 \text{ J K}^{-1} \text{ mol}^{-1}$), T is the temperature (K), N_A is Avogadro's number ($6.022 \times 10^{23} \text{ mol}^{-1}$), A is the electrode area (cm^2), and Γ (molecules nm^{-2}) is the surface density, which can be obtained from the slope of i_p versus v .

Eq. 5 is generally valid for an electrochemically reversible system involving electrode-adsorbed species (SAMs) with homogeneous electron transfer kinetics that are significantly faster than the scan rate.¹⁷ In such an ideal system, the peak-to-peak separation and full width at half maximum (FWHM) are expected to be 0 mV and 90.6 mV, respectively. Additionally, the oxidation and reduction peaks in CVs would exhibit Gaussian shapes, with peak currents directly proportional to the scan rate.^{16,17} However, in our study, CV peaks exhibit a nonzero peak-to-peak separation and an increased FWHM ($>90.6 \text{ mV}$), indicating a quasi-reversible behavior of SAMs on the electrode. This non-ideal reversibility may arise from intermolecular interactions among SAMs as well as from inhomogeneous SAM distribution, where individual SAMs experience different local environments due to neighboring molecules or surrounding electrolytes.^{18,19} These factors cause variations in charge transfer kinetics between SAMs and the electrode, leading to increased FWHM and peak-to-peak separation.

To obtain a more reliable interpretation of SAM coverage using **Eq. 5**, we first performed Gaussian peak deconvolution of the oxidative/reductive peaks in the CVs for different SAMs. For MeO-2PACz, the oxidation peak was best fitted by two subpeaks, suggesting the presence of surface-bound SAMs in two distinct chemical environments. For 2PACz, only one subpeak was needed. For 4PADCB, we analyzed the reductive peak instead of the oxidative peak because, within the selected potential window, the baseline for the oxidative peak could not be reliably determined, whereas the reductive peak allowed for more consistent baseline extraction. Similar to 2PACz, for 4PADCB, only one subpeak was needed for fitting. Finally, we applied **Eq. 5** to calculate the coverage for SAMs assigned to Peak 1 and Peak 2 (if needed). The sum of these two values (if necessary) was then used to represent the total SAM coverage for all samples in this study.

Scanning electron microscopy (SEM) images

SEM images of the prepared perovskite thin films were captured using a TECAN MIRA3 FEG SEM at the Michigan Center for Materials Characterization. Grain size analysis was performed using ImageJ.

Water contact angle measurements

A goniometer (Theta Lite, Biolin-Scientific) was used to measure the water-in-air contact angles on the sample substrates. 5 μ l of water droplet was initially deposited on the substrate, and the contact angle was measured after equilibrating the droplet for 15 s under an ambient atmosphere. The program Attension (Biolin-Scientific) was used to analyze the three-phase contact angle data.

Mechanical characterization

The double cantilever beam (DCB) samples adopted a structure of glass/ITO/SAM with different treatments/perovskite/PMMA/Ag/epoxy/glass. The PMMA layer (~800 nm) was spin-coated onto the perovskite for protection against the Ag, using a 0.1 g/mL PMMA solution in chlorobenzene at 3000 rpm for 30 s. The epoxy used was Loctite epoxy instant mix 5min, which was applied to a bare glass substrate and bonded to the glass substrate deposited with perovskite stacks to form DCB samples. Specific details of the testing method can be found elsewhere.²⁰ Briefly, a pre-crack was introduced into the DCB sample along the orientation of the crack propagation by inserting the tip of a razor blade into the DCB sample. The resulting pre-crack can protect the DCB sample from excessive tensile loads and initiate crack propagation. By using the delaminator system (DTS, USA), the pre-cracked DCB sample was mounted and loaded in tension at a displacement rate of 1 μ m/s. When a clear drop occurred in the load-displacement curve, the DCB sample was partially unloaded to calculate the compliance and then loaded again to gradually propagate the crack until a complete separation was achieved for the two bonded glass substrates, suggesting the completion of the measurement. In the measurement, the load (P) – displacement (Δ) curves were continuously recorded and used to extract the fracture energy (G_c), which can be calculated from **Eq. 6**.²¹

$$G_c = \frac{12 P_c^2 a^2}{B^2 E' h^3} \left(1 + 0.64 \frac{h}{a} \right)^2 \quad (\text{Eq. 6})$$

where P_c is the critical load that deviates from the linearity in the P - Δ curve during the loading cycle; a is the crack length; B and h are the width and half height of the sample, respectively; and

E' is the plane-strain elastic modulus of the substrate. The crack length was estimated using the compliance method in **Eq. 7**:

$$a = \left(\frac{d\Delta}{dP} * \frac{BE'h^3}{8} \right)^{\frac{1}{3}} - 0.64 * h \text{ (Eq. 7)}$$

All G_c testing was carried out in laboratory air environment at $\sim 20^\circ\text{C}$ and $\sim 20\%$ R.H.

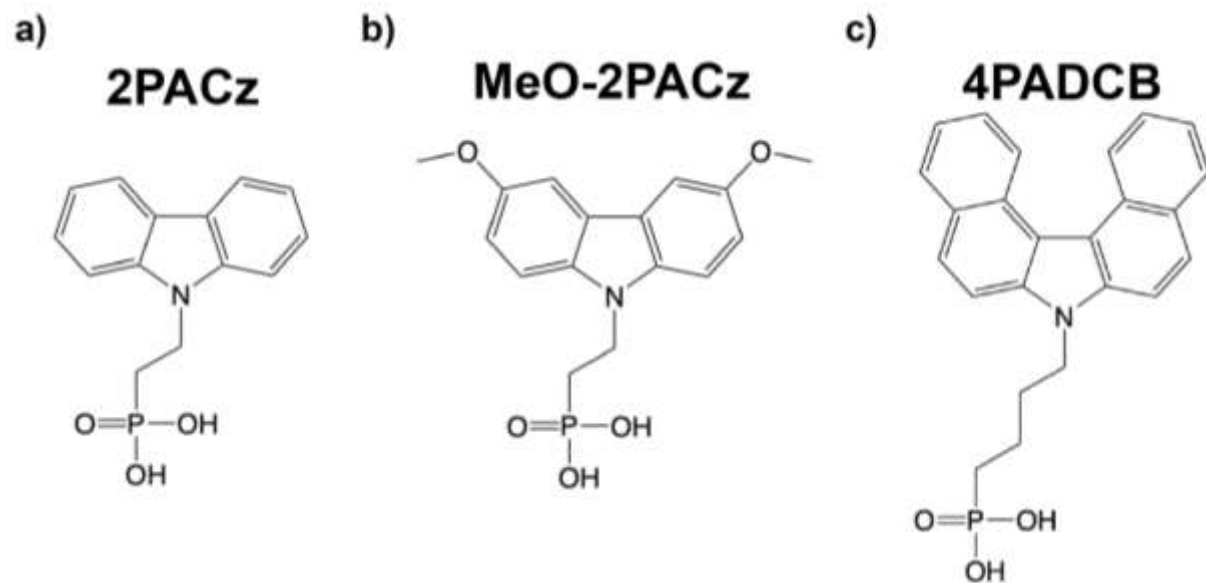


Figure S1: Chemical structures of 2PACz, MeO-2PACz, and 4PADCB SAMs.

Table S1: Definitions of key terms used throughout the manuscript.

Term	Definition
Covalent Bond	A chemical bond resulting from the sharing of an electron pair between two atoms.
Hydrogen Bond	A type of dipole–dipole interaction between a hydrogen atom covalently bonded to a highly electronegative atom (e.g., oxygen, nitrogen, or fluorine) and another highly electronegative atom possessing a lone pair of electrons.
Chemisorption	An adsorption process involving a chemical reaction between the adsorbate and the surface. The interaction can occur through various bonding types depending on the system (often covalent in nature) and is typically characterized by higher adsorption energies than species adsorbed non-reactively.
Physisorption (Weakly Bound)	Adsorption onto a surface through relatively weak, non-covalent interactions such as van der Waals forces or hydrogen bonding, generally resulting in weaker adsorption and less stable attachment compared to chemisorption. In this work, molecules that are viewed as physisorbed/weakly bound include those that are (i) hydrogen bonded, (ii) adsorbed via the carbazole group (i.e., lying flat on the ITO substrate), and (iii) in secondary layers.
First Layer	A term used to describe SAMs within the first monolayer between the ITO substrate and the overall SAM structure. Molecules within the first layer are likely to be directly adsorbed to the substrate, either via chemisorption or physisorption, and detectable by CV, given that the SAMs are (i) electrochemically active molecules, and (ii) either directly adsorbed to the substrate or within a suitable distance for electron tunneling.
Secondary Layer/Multilayer	Refers to molecules not within the first monolayer at the SAM/ITO interface. Molecules in these layer(s) are considered physisorbed—bound indirectly via non-covalent interactions (e.g., π – π stacking or van der Waals forces) with molecules below or adjacent to them rather than by direct adsorption to the substrate. These molecules are detectable by XPS but not by CV.

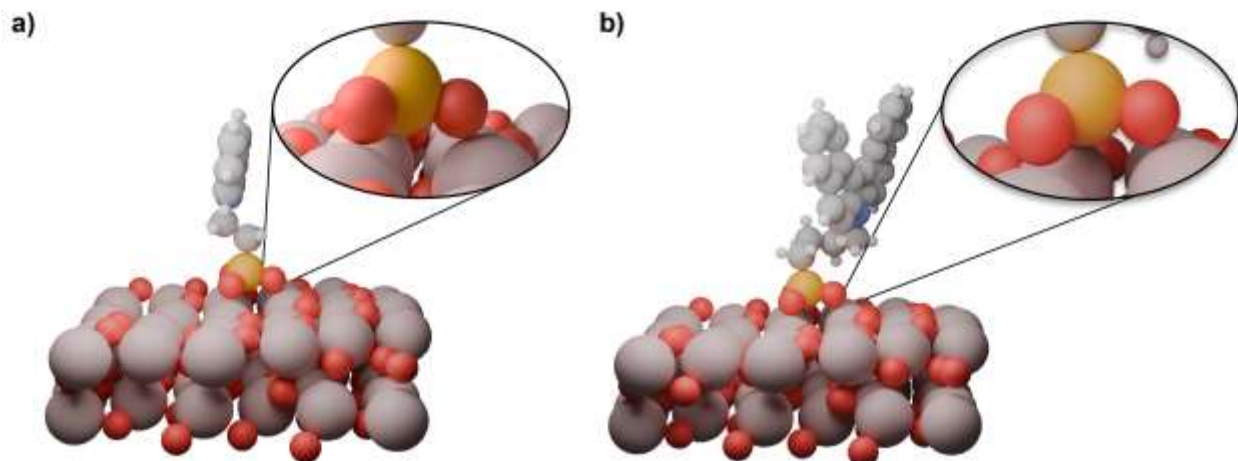


Figure S2: DFT-optimized structures of (a) 2PACz and (b) 4PADCB bound to $\text{In}_2\text{O}_3(111)$ through reactive adsorption via a heterocondensation reaction. Insets highlight the phosphonic acid regions of the SAM molecules to emphasize deprotonation.

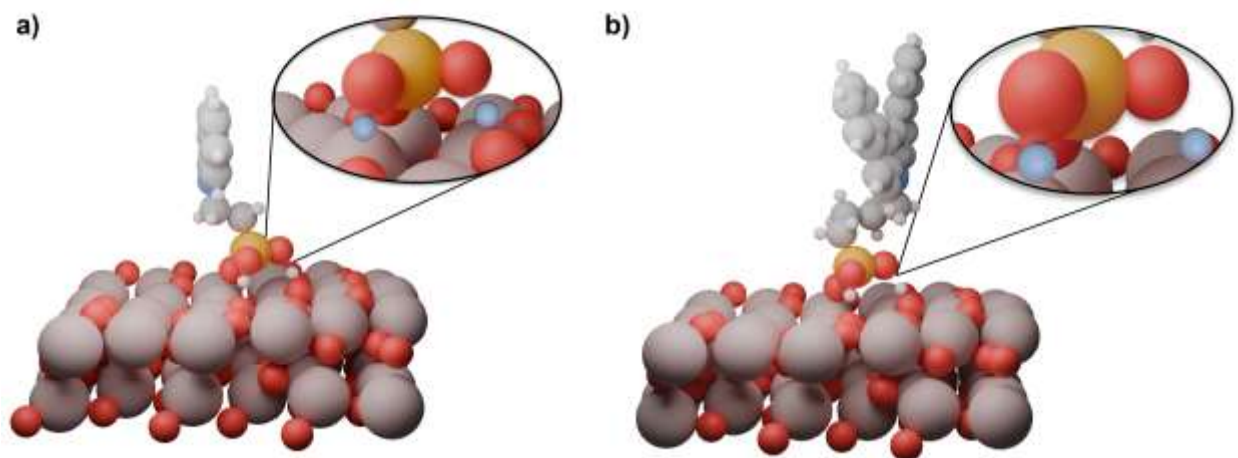


Figure S3: DFT-optimized structures of (a) 2PACz and (b) 4PADCB bound to $\text{In}_2\text{O}_3(111)$ through non-reactive adsorption via hydrogen bonding. Insets highlight the phosphonic acid regions of the SAM molecules to emphasize the lack of deprotonation. Hydrogen atoms in the insets are highlighted in blue to more easily visualize.

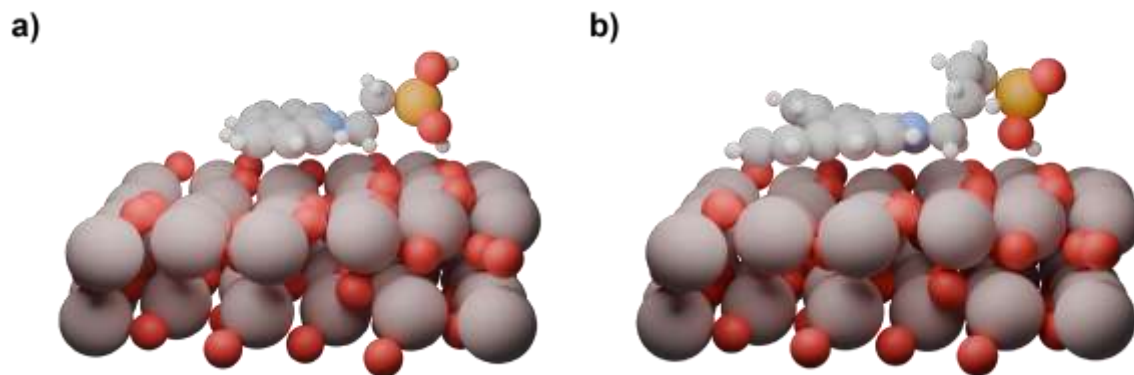


Figure S4: DFT-optimized structures of (a) 2PACz and (b) 4PADCB bound to $\text{In}_2\text{O}_3(111)$ through non-reactive adsorption through the carbazole group.

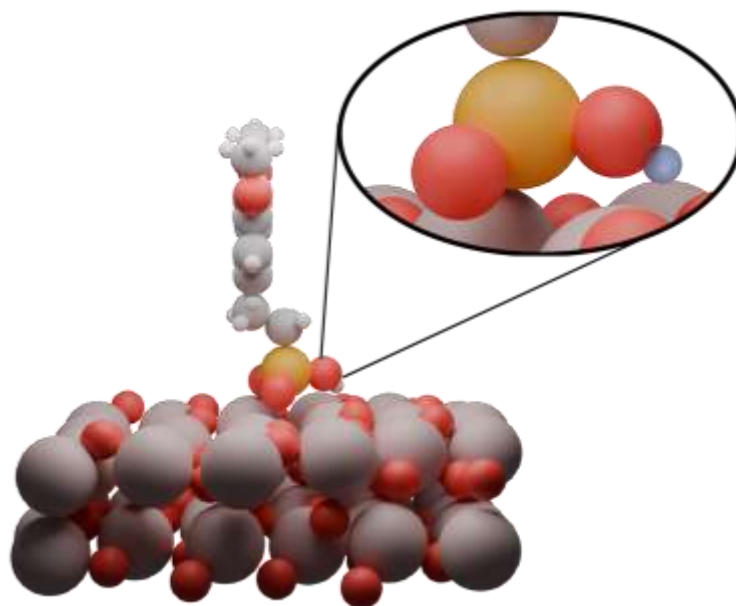


Figure S5: DFT-optimized structure of MeO-2PACz bound to $\text{In}_2\text{O}_3(111)$ via reactive adsorption, forming one covalent bond. The inset highlights the phosphonic acid region of MeO-2PACz to emphasize which group is deprotonated. Hydrogen atoms in the inset are highlighted in blue to facilitate easier visualization.

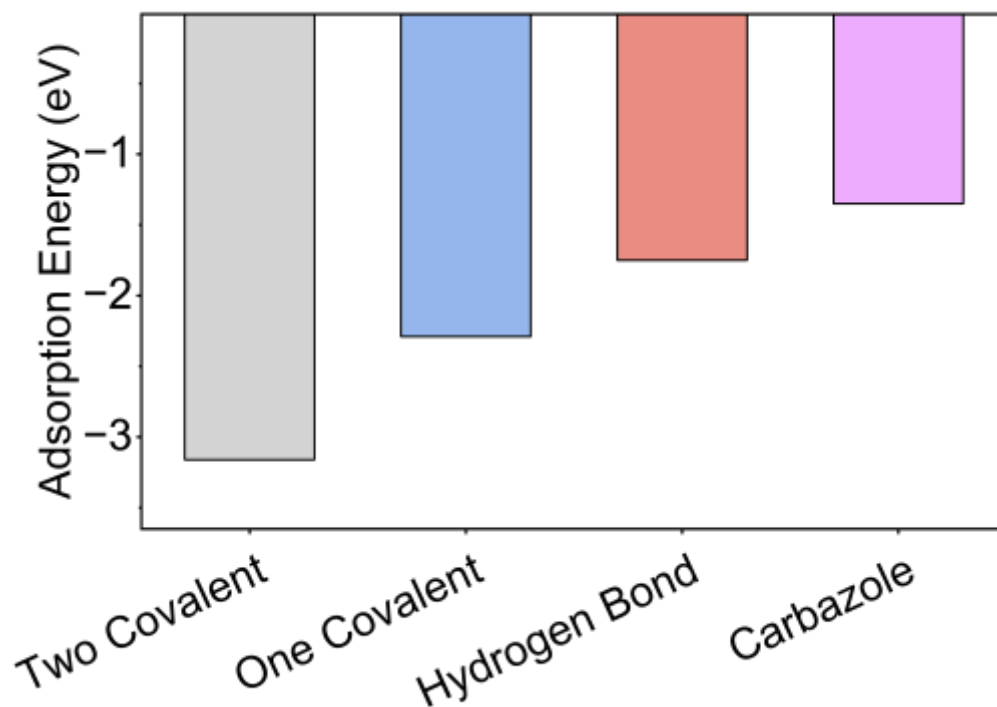


Figure S6: DFT-calculated adsorption energies for each mechanism for MeO-2PACz, shown left to right as reactive adsorption (two covalent bonds versus one covalent bond), hydrogen bonding, and carbazole-based adsorption, respectively. The reactive adsorption energies were computed following the deprotonation step and water formation to directly quantify the strength of the chemisorbed SAM molecules.

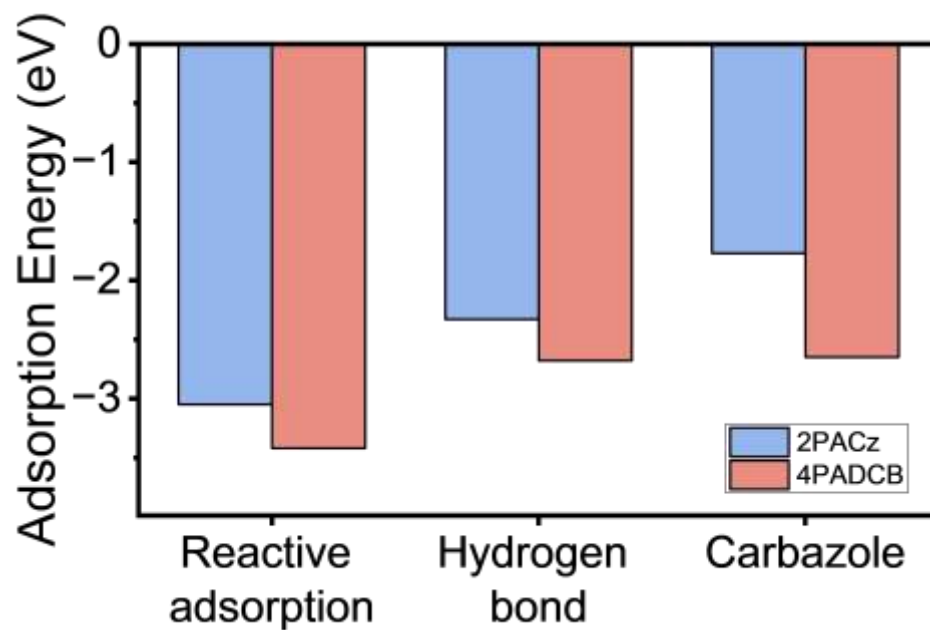


Figure S7: DFT-calculated adsorption energies for each mechanism for 2PACz and 4PADCB, shown left to right as reactive adsorption, hydrogen bonding, and carbazole-based adsorption, respectively. The reactive adsorption energy was computed following the deprotonation step and water formation to directly quantify the strength of the chemisorbed SAM molecule.

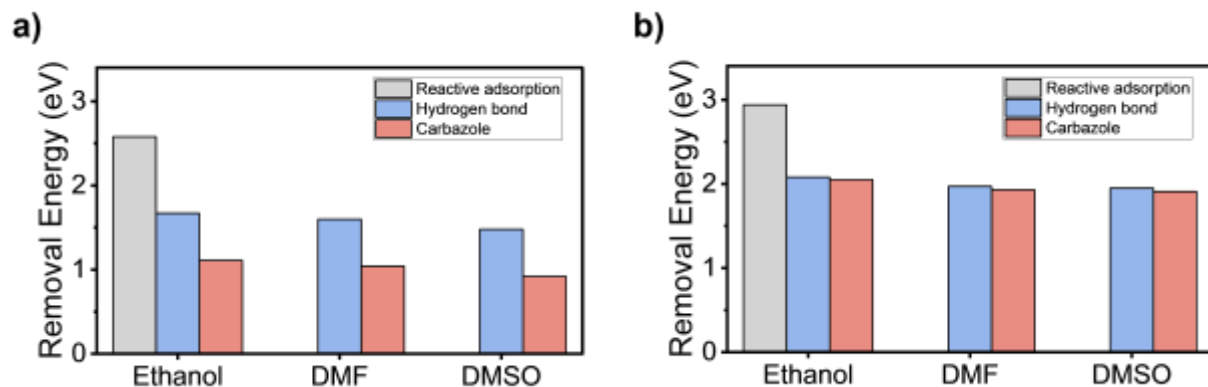


Figure S8: DFT-calculated removal energies of adsorbed (a) 2PACz and (b) 4PADCB by ethanol, DMF, and DMSO. Bars represent removal energies for SAM molecules adsorbed via reactive adsorption (red), hydrogen bonding (blue), and the carbazole group (gray). Due to the lack of hydrogen bonding between DMF or DMSO and the reactively adsorbed SAM molecules, both solvents are assumed to be ineffective at displacing reactively adsorbed SAM molecules.

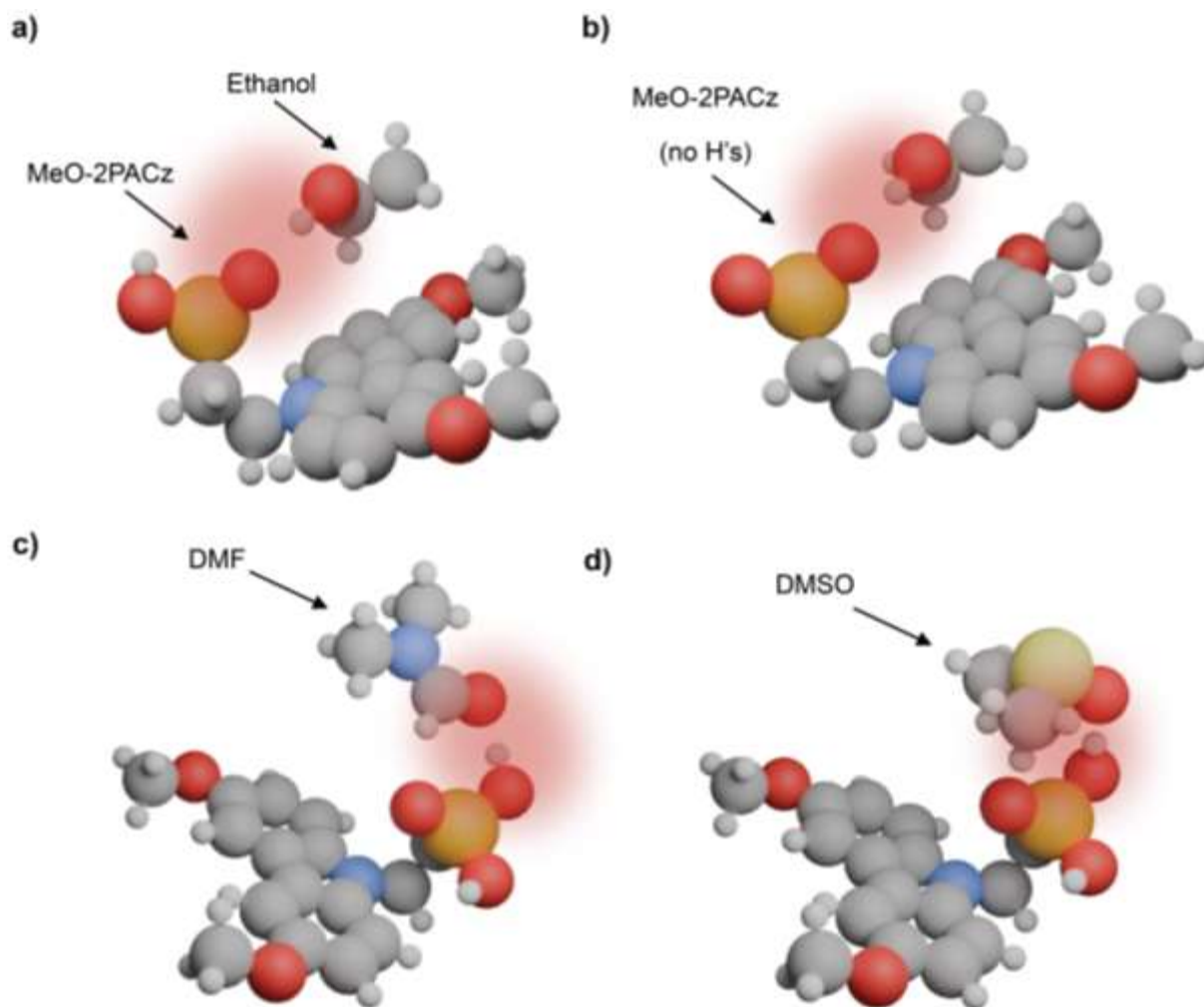


Figure S9: DFT optimized structures of (a) ethanol bound to MeO-2PACz, (b) ethanol bound to a deprotonated MeO-2PACz, (c) DMF bound to MeO-2PACz, and (d) DMSO bound to MeO-2PACz. Atom color legend: Gray = Carbon; Red = Oxygen; Orange = Phosphorus; Beige = Hydrogen; Yellow = Sulfur; Blue = Nitrogen.

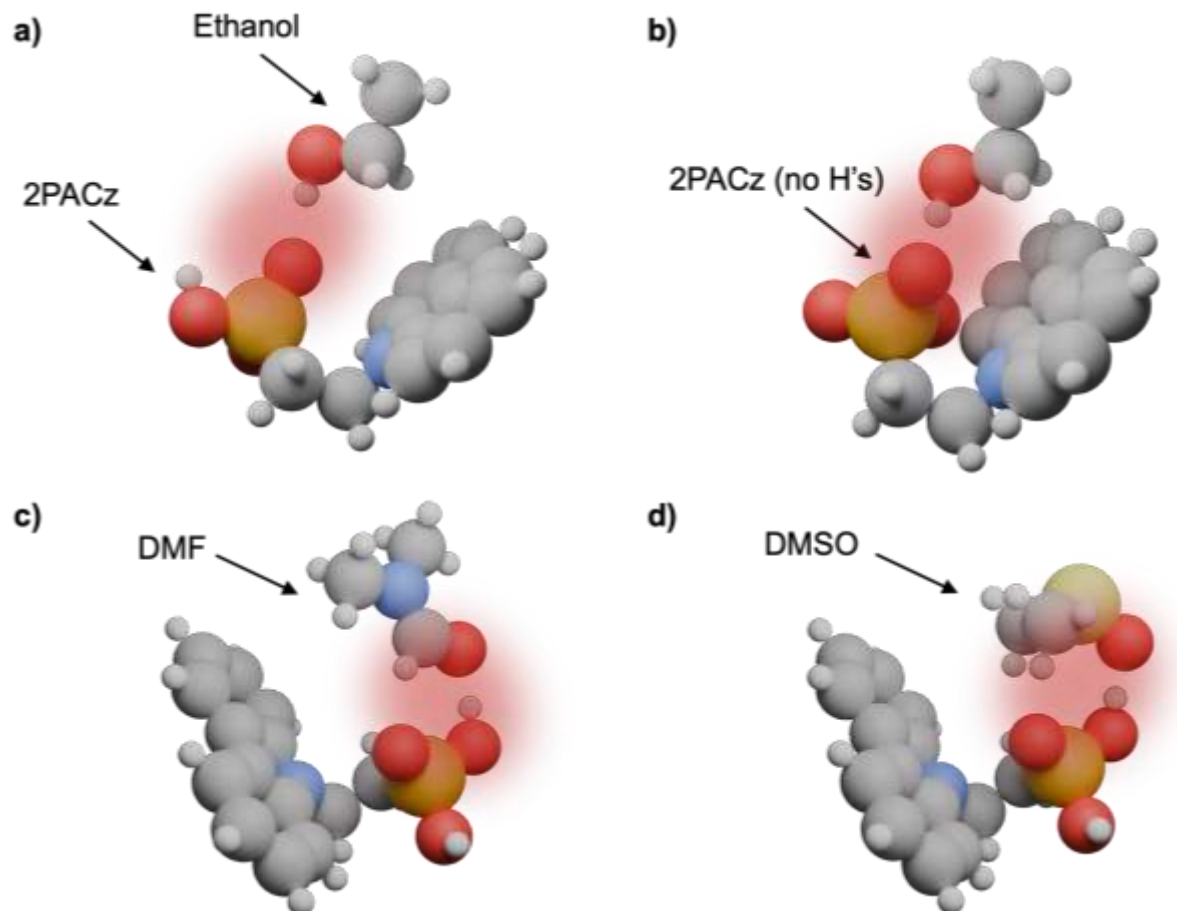


Figure S10: DFT optimized structures of (a) ethanol bound to 2PACz, (b) ethanol bound to a deprotonated 2PACz, (c) DMF bound to 2PACz, and (d) DMSO bound to 2PACz. Atom color legend: Gray = Carbon; Red = Oxygen; Orange = Phosphorus; Beige = Hydrogen; Yellow = Sulfur; Blue = Nitrogen.

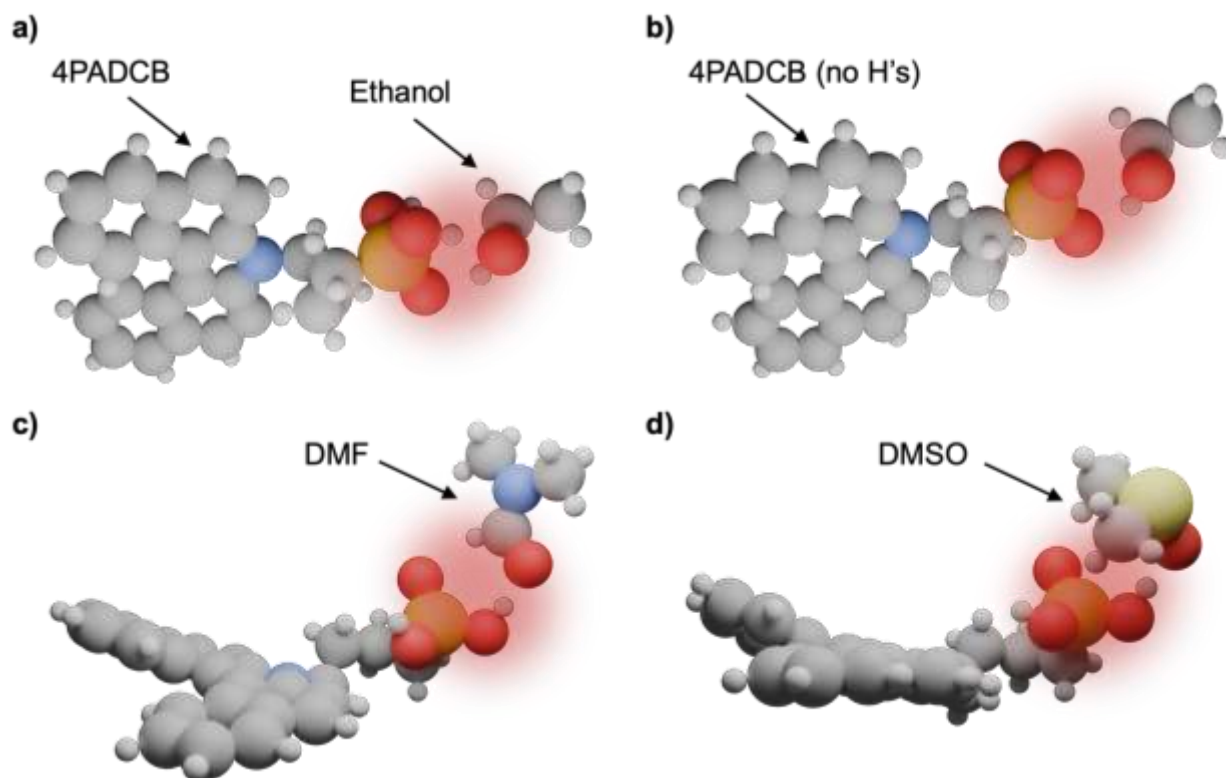


Figure S11: DFT optimized structures of (a) ethanol bound to 4PADCB, (b) ethanol bound to a deprotonated 4PADCB, (c) DMF bound to 4PADCB, and (d) DMSO bound to 4PADCB. Atom color legend: Gray = Carbon; Red = Oxygen; Orange = Phosphorus; Beige = Hydrogen; Yellow = Sulfur; Blue = Nitrogen.

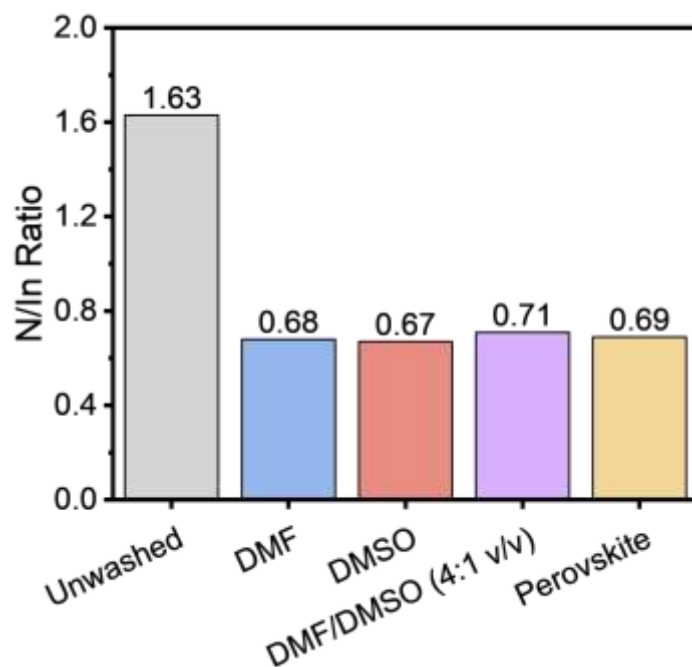


Figure S12: N/In ratio extracted from XPS spectra of MeO-2PACz films on ITO that were, from left to right, unwashed, washed twice with DMF, washed twice with DMSO, washed twice with DMF/DMSO (4:1 v/v), and washed with the perovskite precursor solution ($\text{Cs}_{0.06}\text{FA}_{0.94}\text{PbI}_3$ in DMF/DMSO (4:1 v/v) without antisolvent) followed by a DMF wash to remove residual solutes.

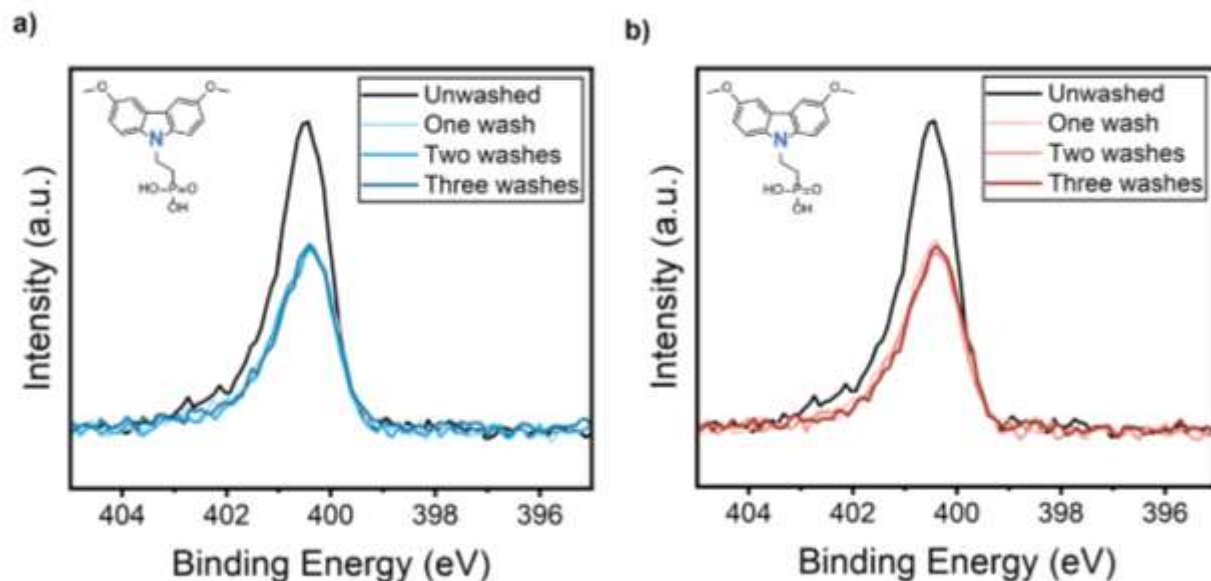


Figure S13: N 1s XPS spectra of MeO-2PACz films that underwent no washing, one washing step, two washing steps, and three washing steps with (a) ethanol and (b) DMF. It was determined that because of negligible decreases in peak area after two washing steps, all SAM molecules that are removable by a given solvent were removed.

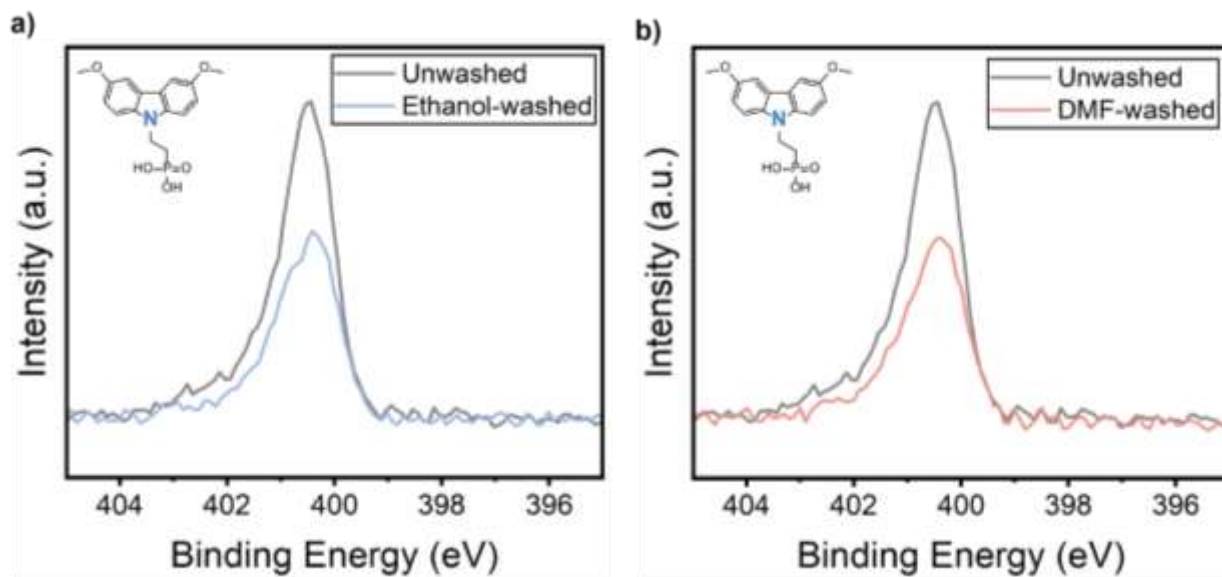


Figure S14: N 1s XPS spectra of MeO-2PACz films before and after two washing steps with (a) ethanol and (b) DMF.

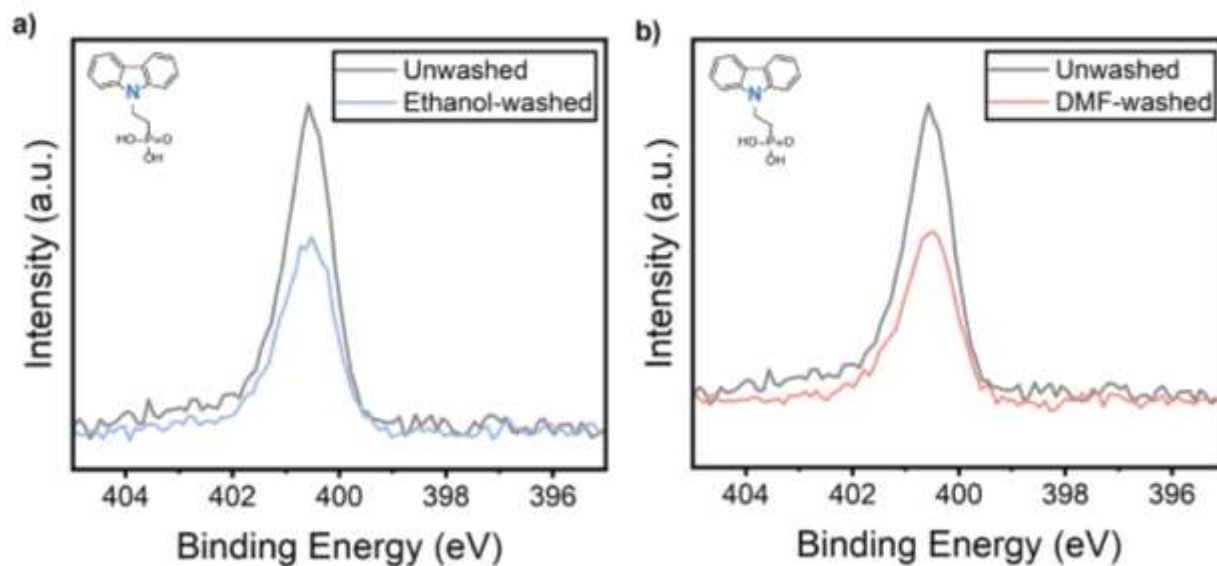


Figure S15: N 1s XPS spectra of 2PACz films before and after two washing steps with (a) ethanol and (b) DMF.

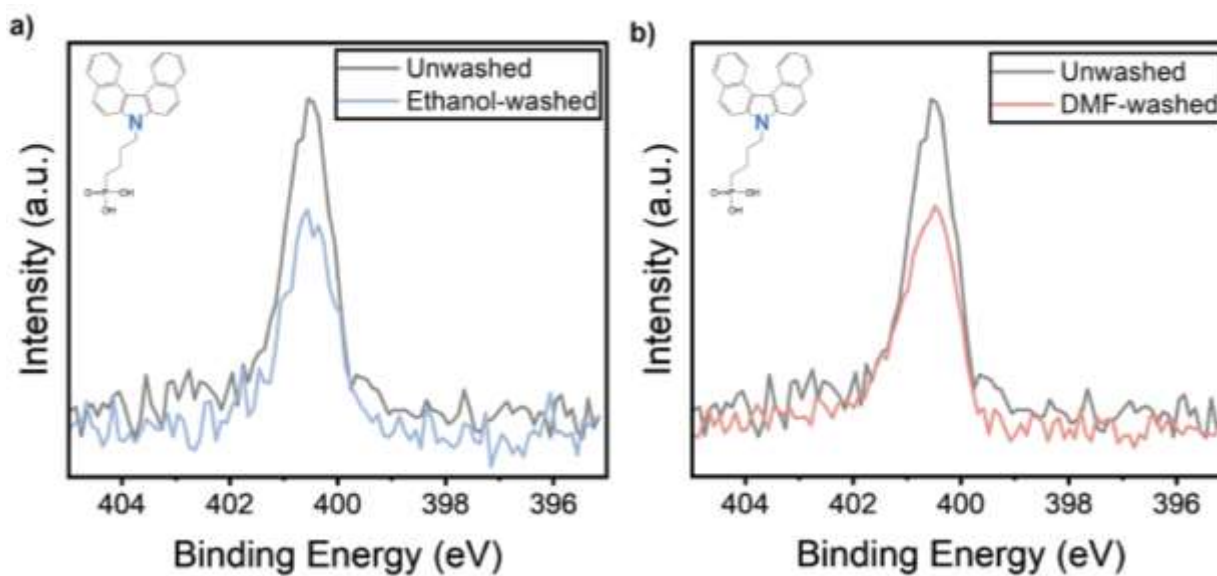


Figure S16: N 1s XPS spectra of 4PADCB films before and after two washing steps with (a) ethanol and (b) DMF.

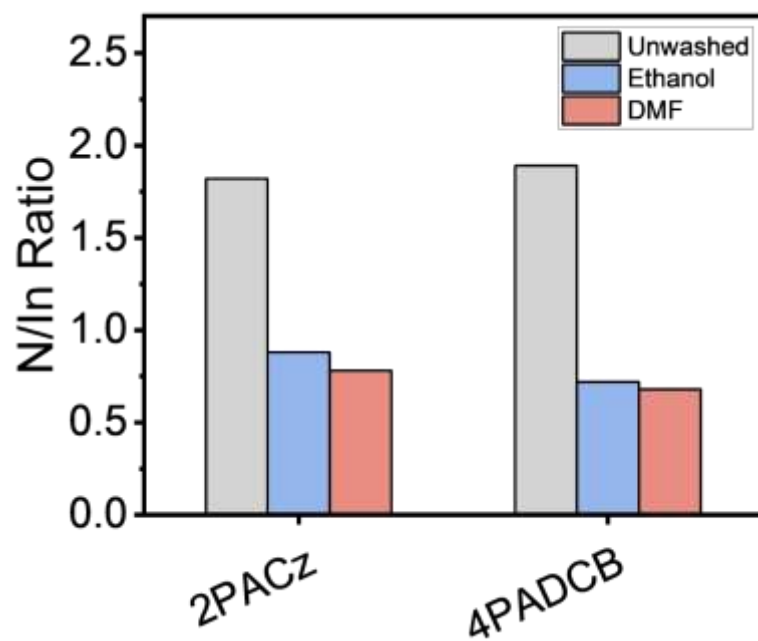


Figure S17: Ratio of nitrogen to indium extracted from XPS spectra of unwashed, twice ethanol-washed, and twice DMF-washed 2PACz and 4PADCB.

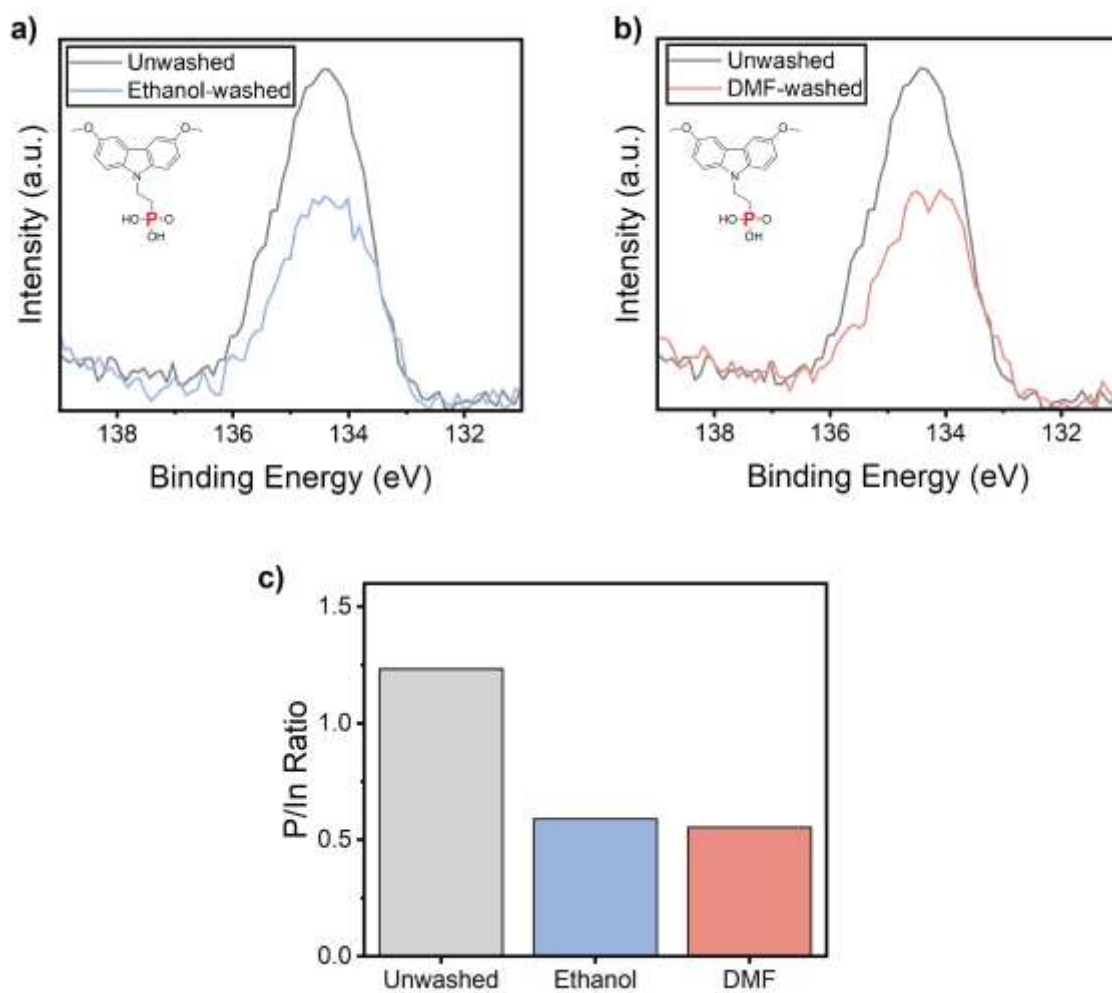


Figure S18: P 2p XPS spectra of MeO-2PACz before and after two washing steps with (a) ethanol and (b) DMF. (c) The ratio of phosphorus to indium of, from left to right, MeO-2PACz, ethanol-washed MeO-2PACz, and DMF-washed MeO-2PACz on ITO.

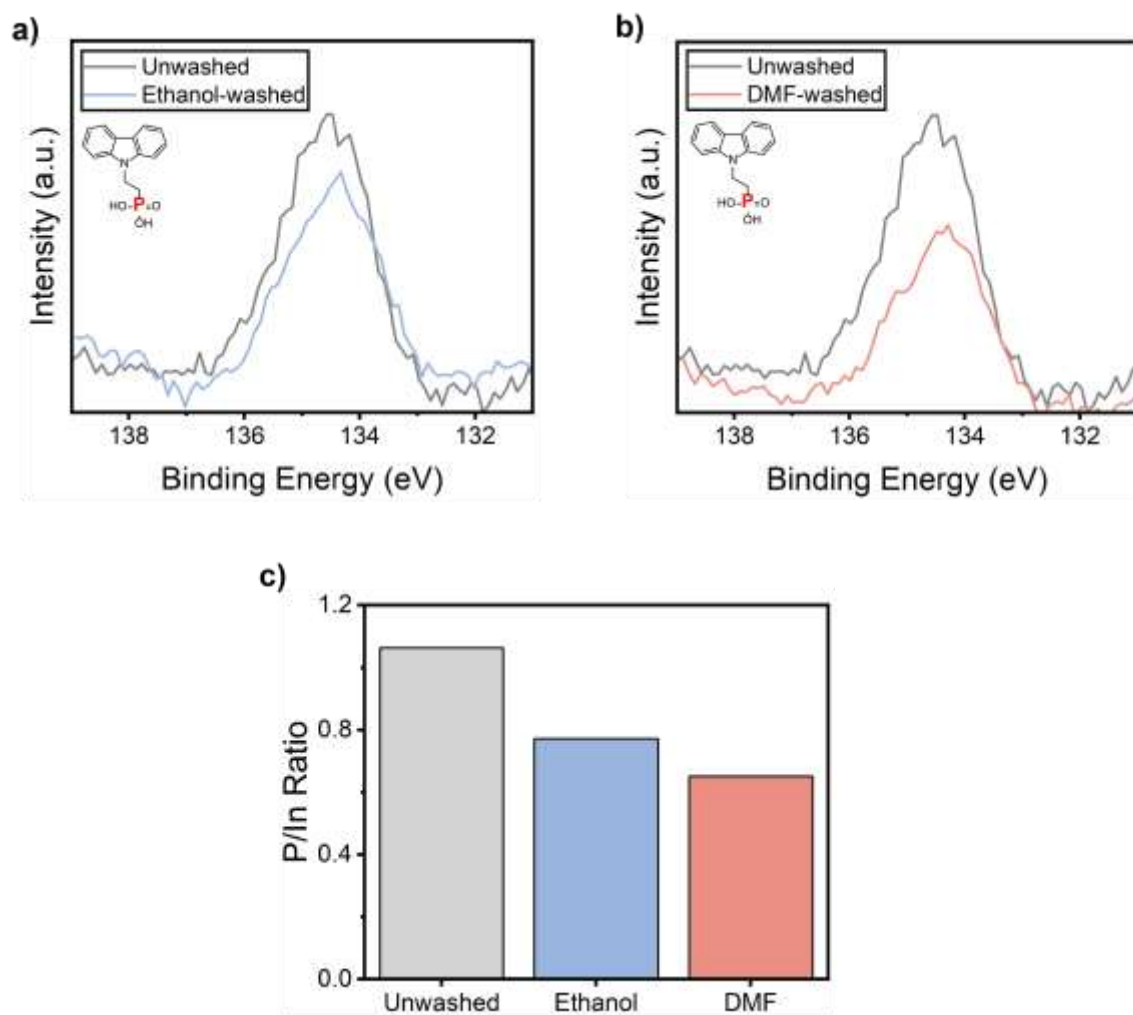


Figure S19: P 2p XPS spectra of 2PACz before and after two washing steps (a) ethanol and (b) DMF. (c) The ratio of phosphorus to indium of, from left to right, unwashed 2PACz, ethanol-washed 2PACz, and DMF-washed 2PACz on ITO.

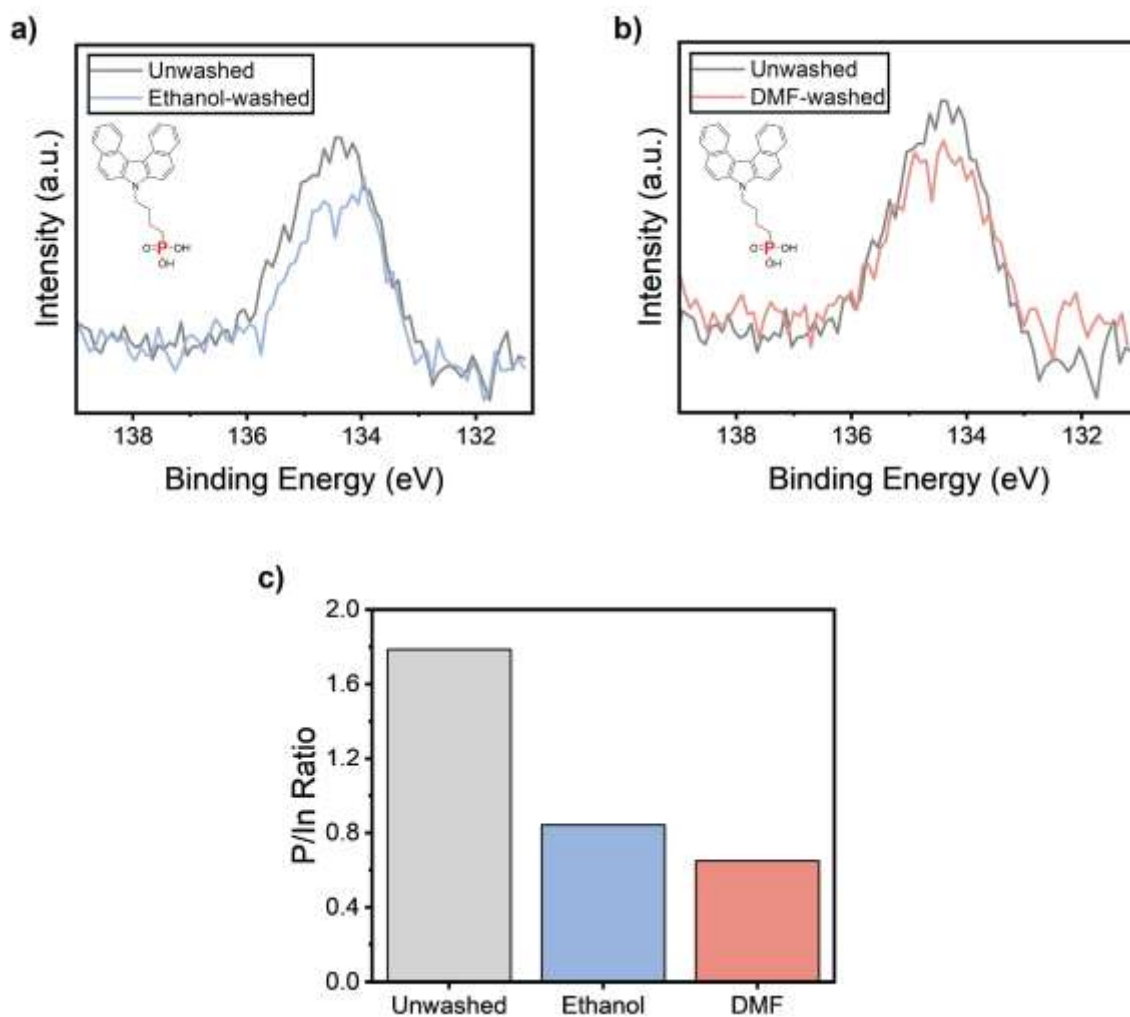


Figure S20: P 2p XPS spectra of 4PADCB before and after two washing steps with (a) ethanol and (b) DMF. (c) The ratio of phosphorus to indium of, from left to right, unwashed 4PADCB, ethanol-washed 4PADCB, and DMF-washed 4PADCB on ITO.

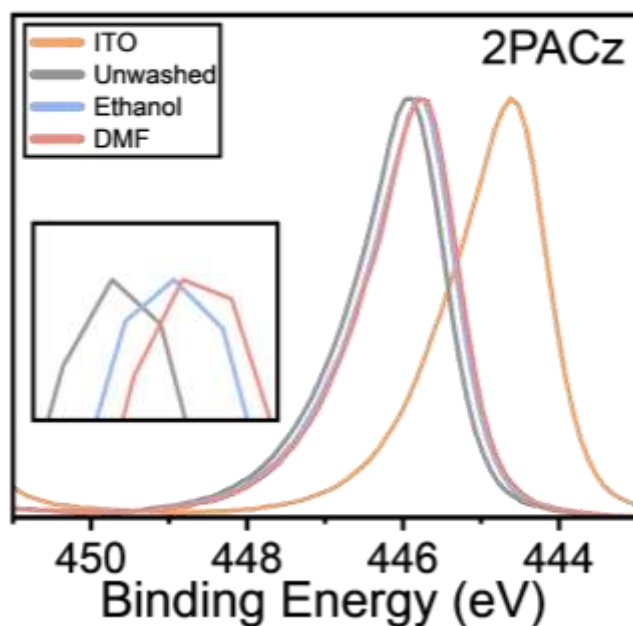


Figure S21: In $3d_{5/2}$ XPS spectra for bare ITO and corresponding unwashed and solvent-washed 2PACz. Peak shifts are highlighted inset.

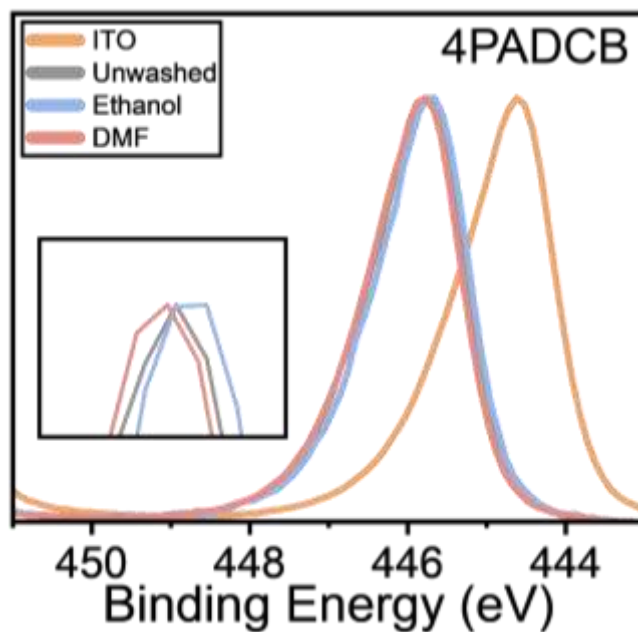


Figure S22: In $3d_{5/2}$ XPS spectra for bare ITO and corresponding unwashed and solvent-washed 4PADCB. Peak shifts are highlighted inset.

Supplementary Note 1: Deposition of 4PADCB produced an initial blue shift of +1.10 eV. Following ethanol washing, we observed a slight red shift of -0.06 eV, similar to MeO-2PACz and 2PACz. However, after DMF washing, rather than a further red shift as seen with MeO-2PACz and 2PACz, the spectrum blue shifted by +0.13 eV, giving a binding energy higher than that of the unwashed sample. We propose that this distinctive behavior may arise from a reconfiguration of 4PADCB molecules after DMF washing. In this scenario, extensive removal of disordered, physisorbed molecules could expose regions of the ITO surface, allowing the remaining chemisorbed 4PADCB molecules to reorganize during annealing into flatter conformations. The flexible 4-carbon alkyl linker and conjugated aromatic moieties may facilitate such a rearrangement, potentially strengthening π -metal interactions with the ITO surface. This reorganization could increase local electron density around surface indium atoms, contributing to the observed blue shift.

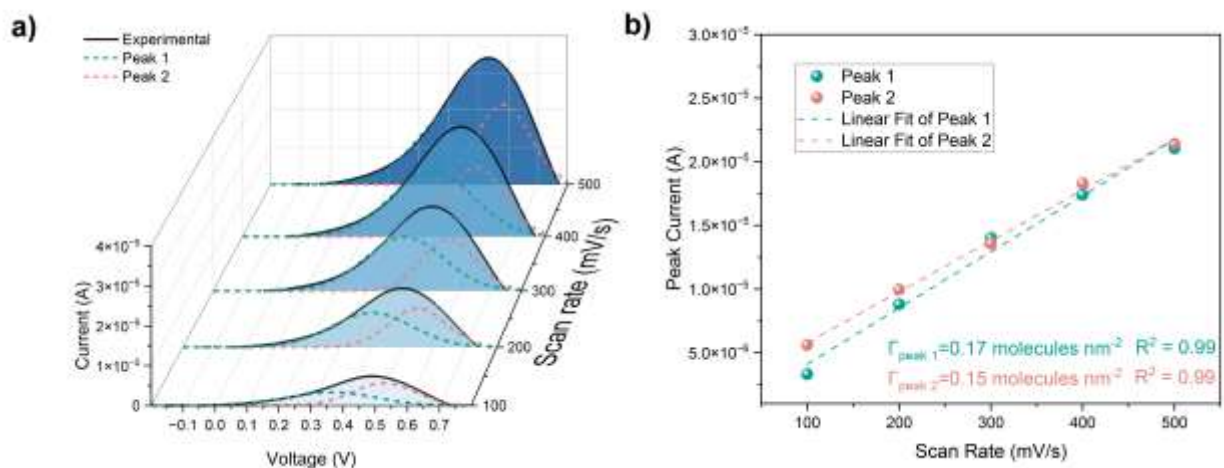


Figure S23: (a) Oxidative peak in cyclic voltammograms of MeO-2PACz-coated ITO substrate before washing, measured at different voltage scan rates. (b) Correlation between the oxidative peak current and the voltage scan rate for Peak 1 and Peak 2. The dashed lines represent linear fits to the extracted peak currents.

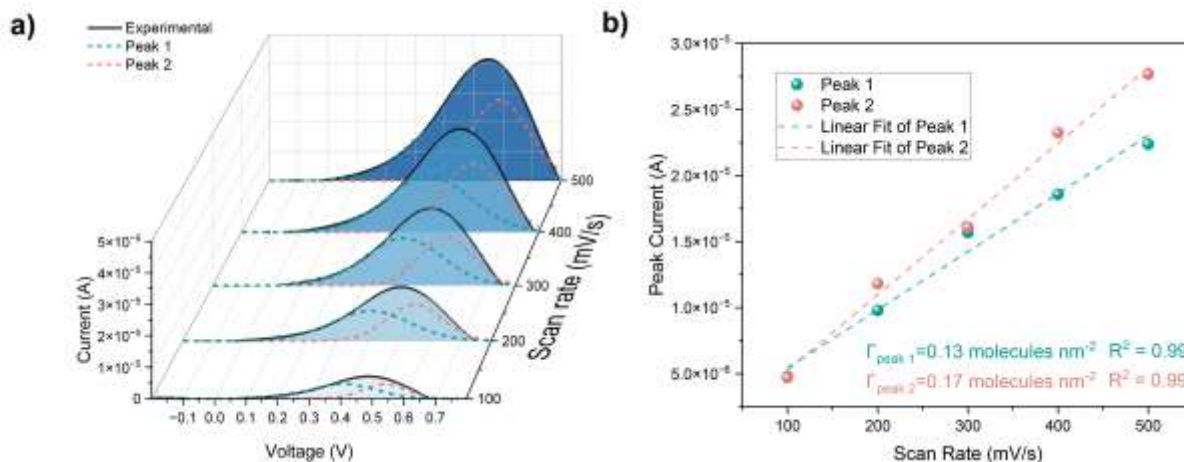


Figure S24: (a) Oxidative peak in cyclic voltammograms of MeO-2PACz-coated ITO substrate after ethanol washing measured at different voltage scan rates. (b) Correlation between the oxidative peak current and the voltage scan rate for Peak 1 and Peak 2. The dashed lines represent linear fits to the extracted peak currents.

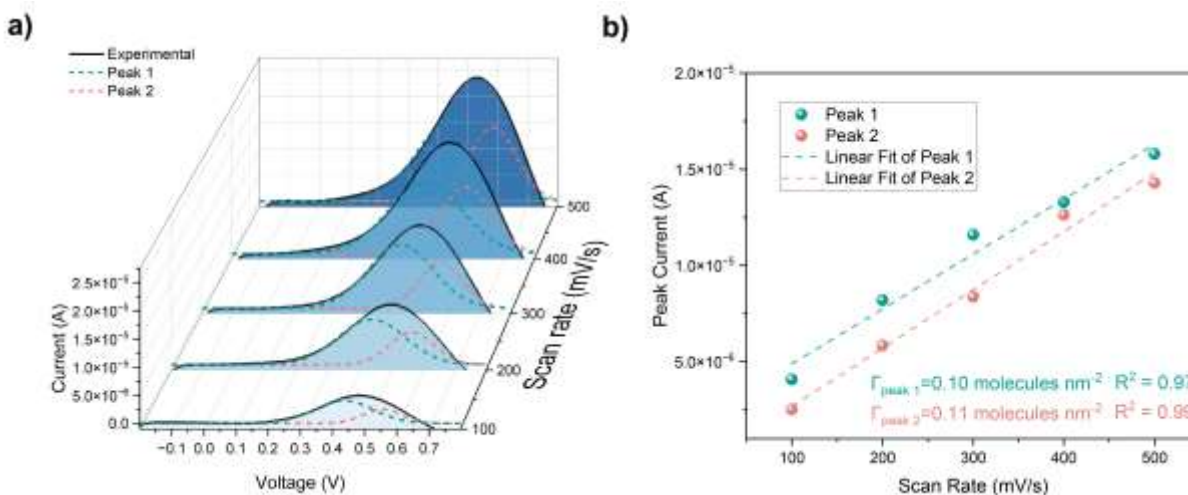


Figure S25: (a) Oxidative peak in cyclic voltammograms of MeO-2PACz-coated ITO substrate after DMF washing measured at different voltage scan rates. (b) Correlation between the oxidative peak current and the voltage scan rate for Peak 1 and Peak 2. The dashed lines represent linear fits to the extracted peak currents.

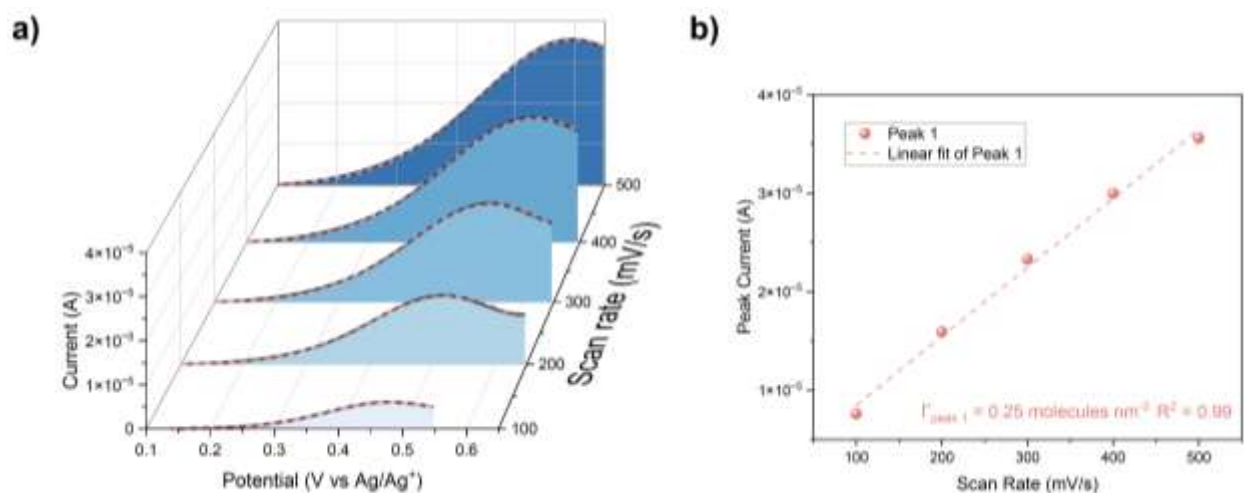


Figure S26: (a) Oxidative peak in cyclic voltammograms of 2PACz-coated ITO substrate before washing, measured at different voltage scan rates. (b) Correlation between the oxidative peak current and the voltage scan rate. The dashed line represents a linear fit to the extracted peak currents.

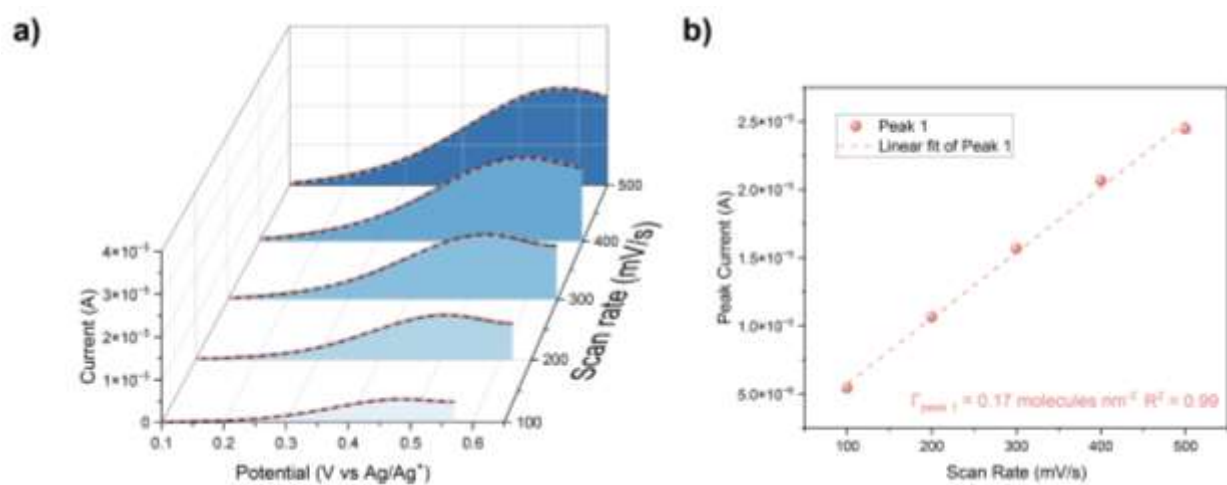


Figure S27: (a) Oxidative peak in cyclic voltammograms of 2PACz-coated ITO substrate after ethanol washing, measured at different voltage scan rates. (b) Correlation between the oxidative peak current and the voltage scan rate. The dashed line represents a linear fit to the extracted peak currents.

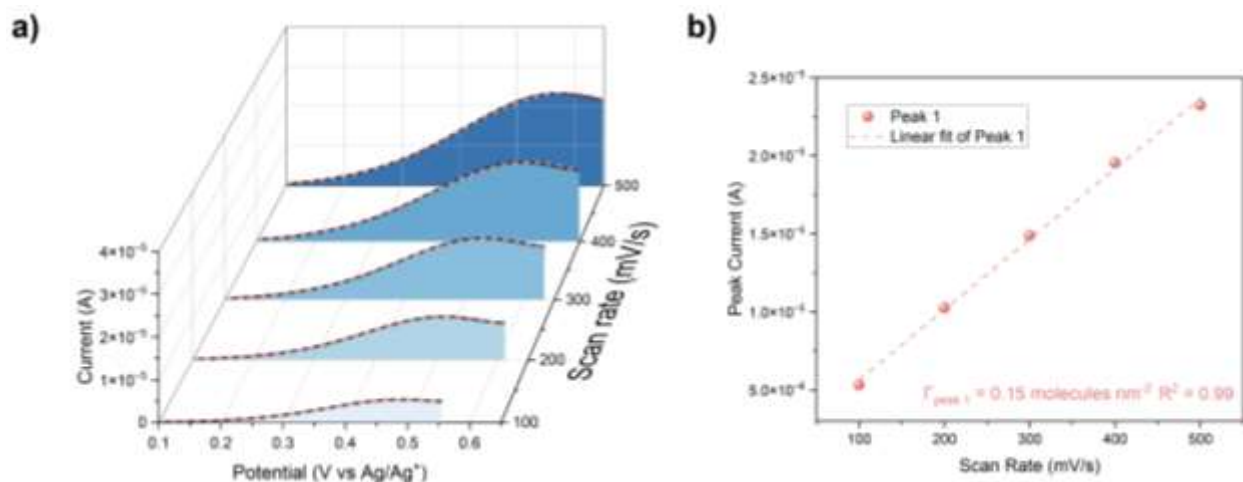


Figure S28: (a) Oxidative peak in cyclic voltammograms of 2PACz-coated ITO substrate after DMF washing, measured at different voltage scan rates. (b) Correlation between the oxidative peak current and the voltage scan rate. The dashed line represents a linear fit to the extracted peak currents.

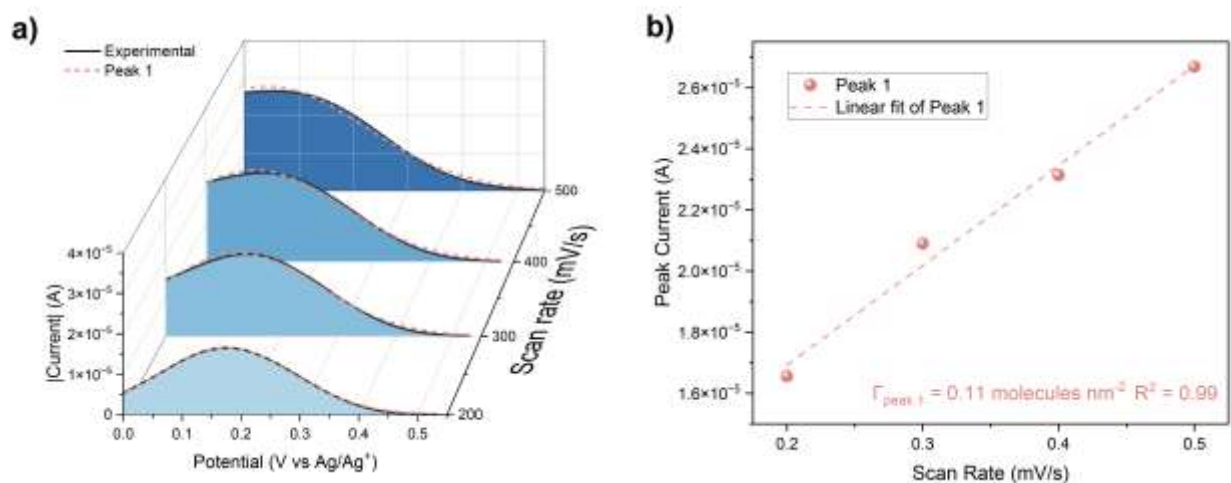


Figure S29: (a) Reductive peak in cyclic voltammograms of 4PADCB-coated ITO substrate before washing, measured at different voltage scan rates. (b) Correlation between the reductive peak current and the voltage scan rate. The dashed line represents a linear fit to the extracted peak currents.

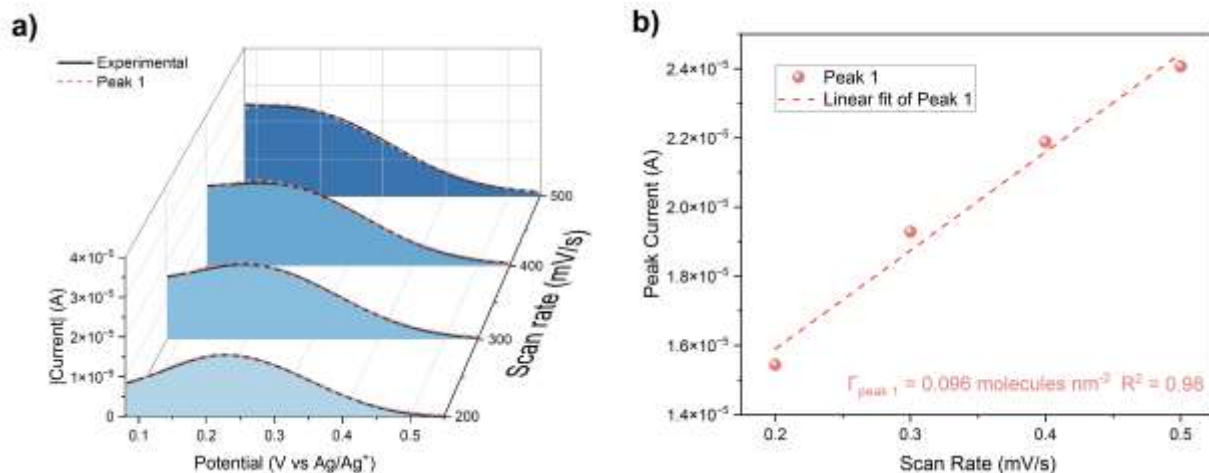


Figure S30: (a) Reductive peak in cyclic voltammograms of 4PADCB-coated ITO substrate after ethanol washing measured at different voltage scan rates. (b) Correlation between the reductive peak current and the voltage scan rate. The dashed line represents a linear fit to the extracted peak currents.

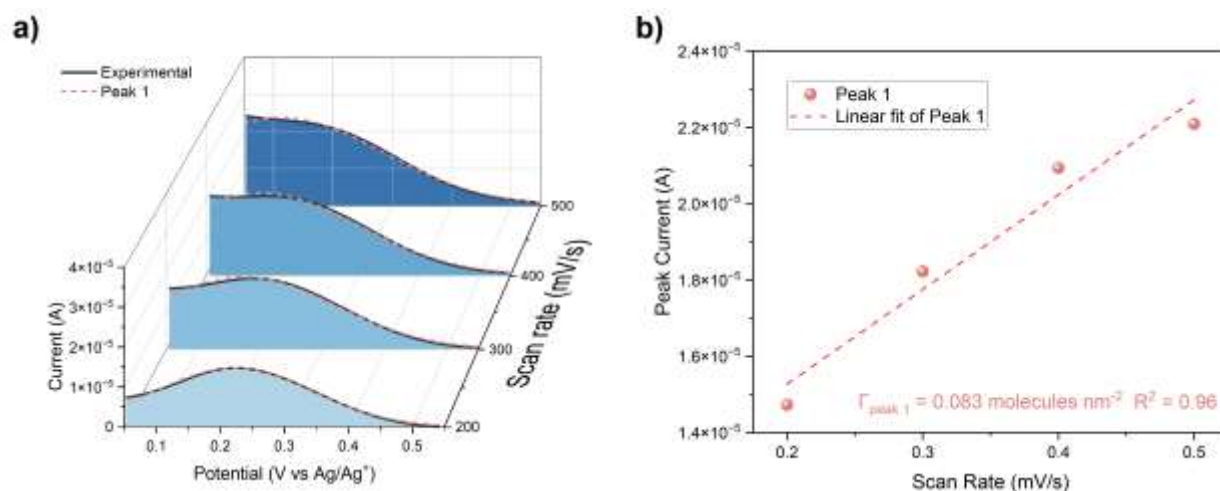


Figure S31: (a) Reductive peak in cyclic voltammograms of 4PADCB-coated ITO substrate after DMF washing measured at different voltage scan rates. (b) Correlation between the reductive peak current and the voltage scan rate. The dashed line represents a linear fit to the extracted peak currents.

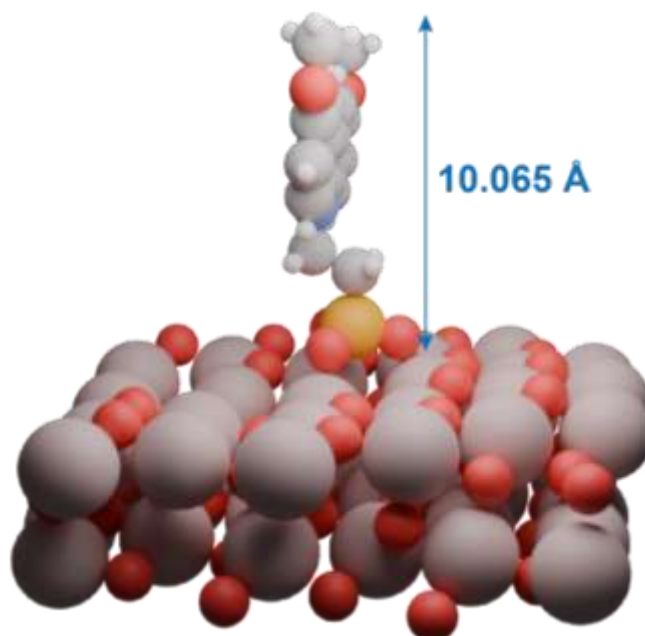


Figure S32: DFT-calculated molecular length of a reactively adsorbed MeO-2PACz molecule on $\text{In}_2\text{O}_3(111)$.

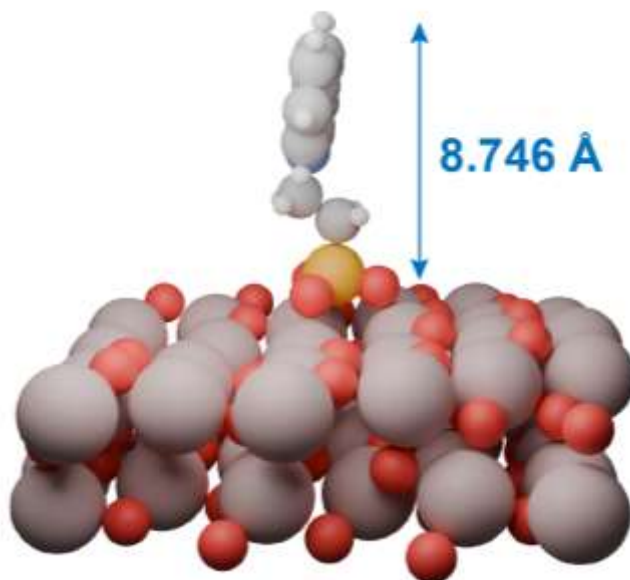


Figure S33: DFT-calculated molecular length of a reactively adsorbed 2PACz molecule on $\text{In}_2\text{O}_3(111)$.

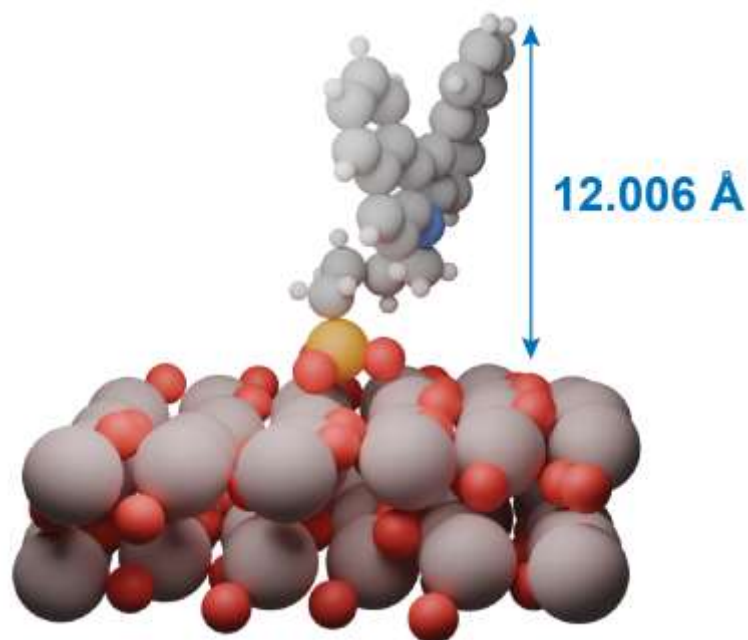


Figure S34: DFT-calculated molecular length of a reactively adsorbed 4PADCB molecule on $\text{In}_2\text{O}_3(111)$.

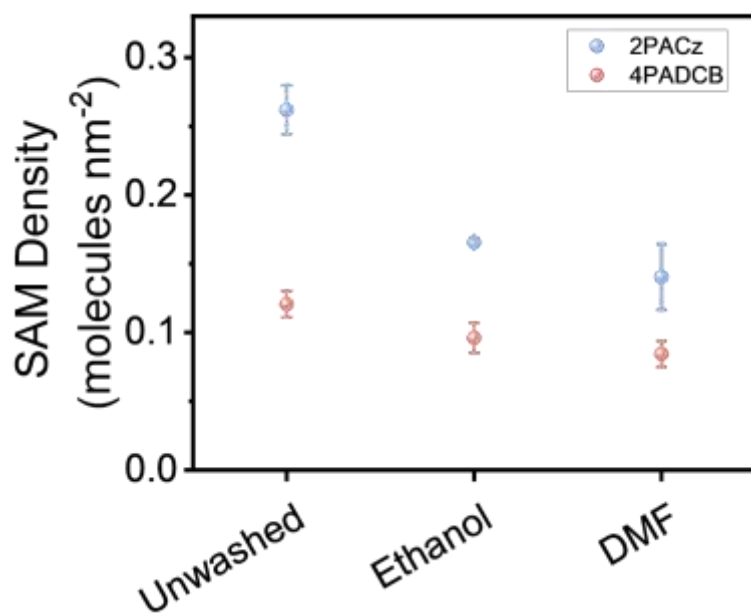


Figure S35: Areal density of unwashed, ethanol-washed, and DMF-washed (a) 2PACz and (b) 4PADCB extracted from CV.

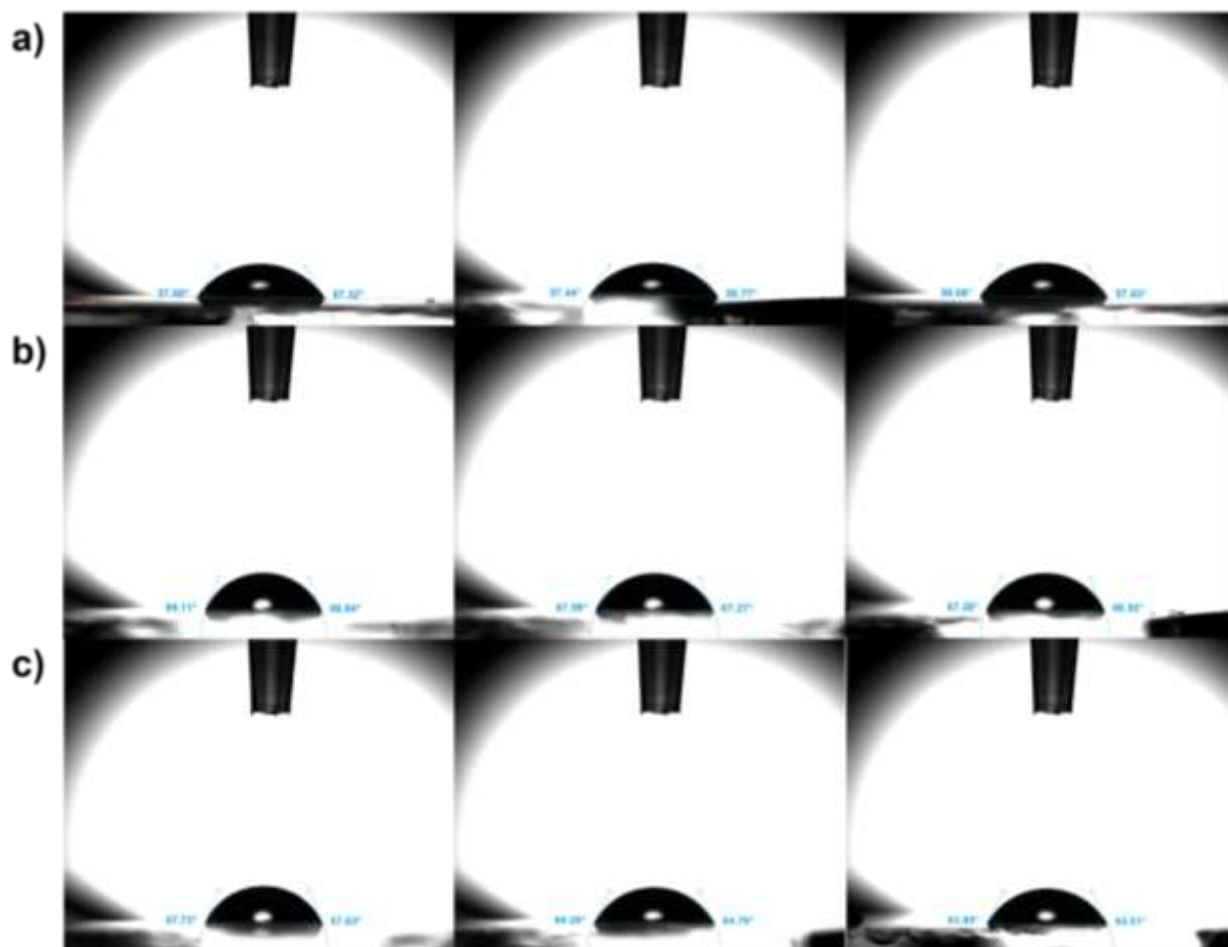


Figure S36: Water contact angle measurements of (a) Unwashed MeO-2PACz-coated ITO substrates (control), (b) MeO-2PACz-coated ITO substrates washed with ethanol, and (c) MeO-2PACz-coated ITO substrates washed with DMF. Each sample had three measurements taken, with each image corresponding to a separate measurement.

Table S2: Water contact angle values on MeO-2PACz films shown in each image of **Fig. S36**. Each sample had three separate measurements taken, with two contact angle values per measurement.

Measurement Name	Value One (°)	Value Two (°)
Unwashed_1	57.60	57.32
Unwashed_2	57.44	56.77
Unwashed_3	58.08	57.83
Ethanol_1	69.11	68.84
Ethanol_2	67.59	67.27
Ethanol_3	67.28	66.92
DMF_1	67.73	67.63
DMF_2	65.29	64.79
DMF_3	63.85	63.61

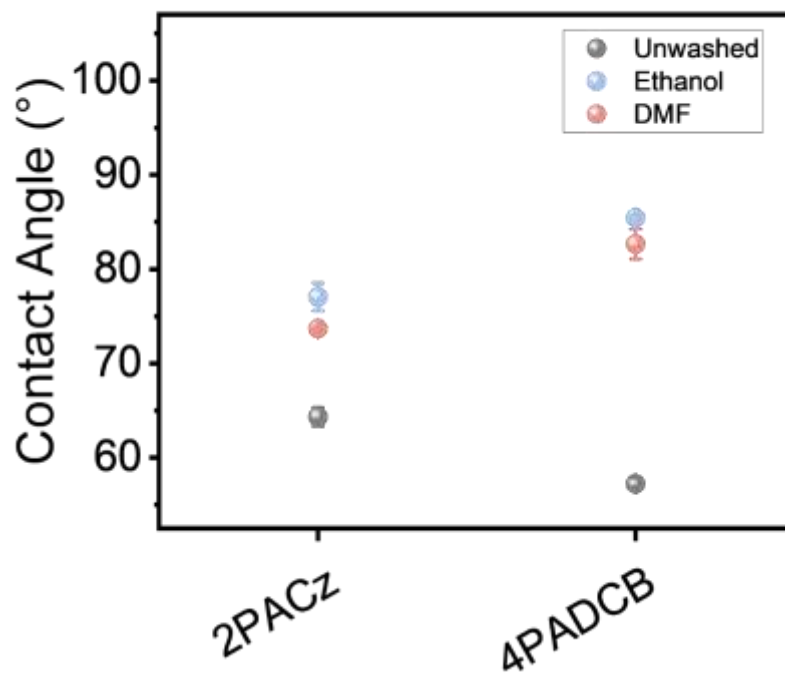


Figure S37: Average water contact angle of unwashed, ethanol-washed, and DMF-washed 2PACz and 4PADCB.

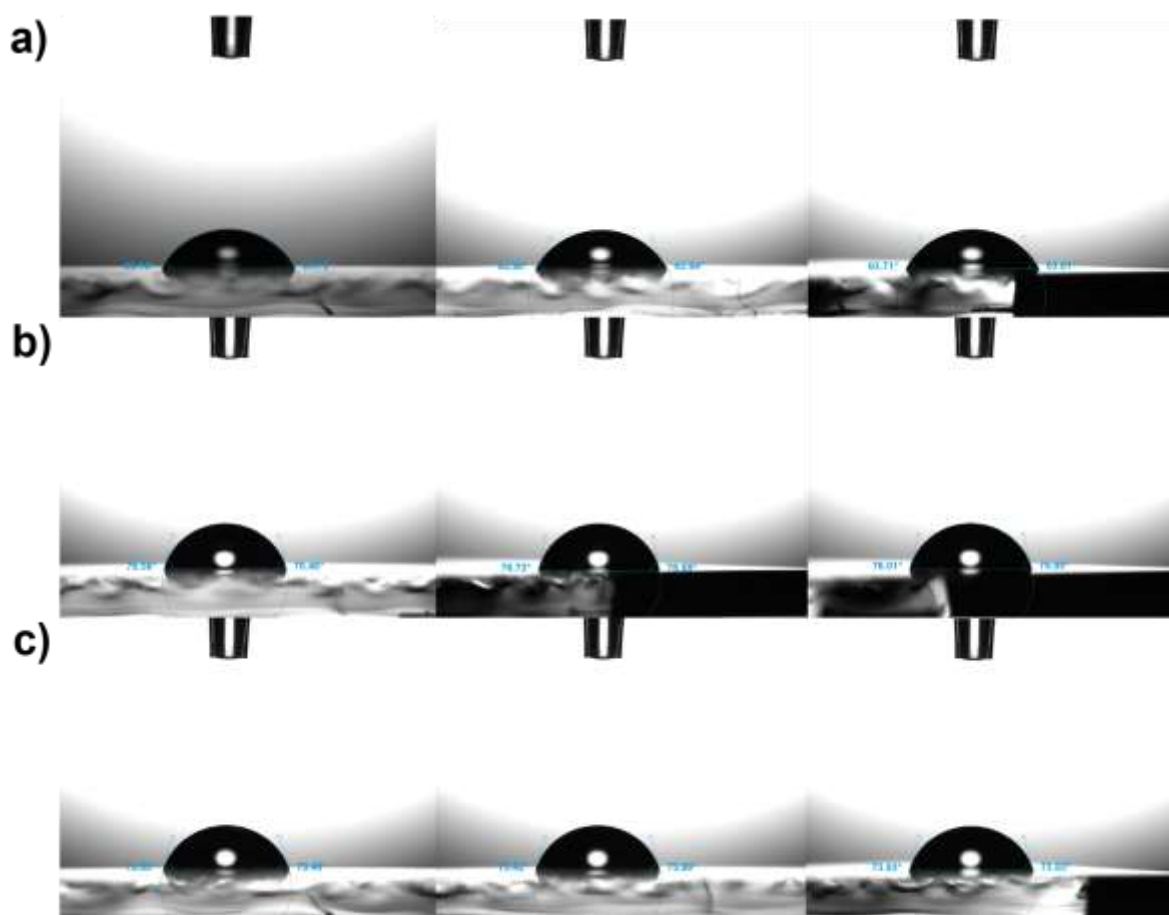


Figure S38: Water contact angle measurements of (a) Unwashed 2PACz-coated ITO substrates (control), (b) 2PACz-coated ITO substrates washed with ethanol, and (c) 2PACz-coated ITO substrates washed with DMF. Each sample had three measurements taken, with each image corresponding to a separate measurement.

Table S3: Water contact angle values on 2PACz films shown in each image of **Fig. S38**. Each sample had three separate measurements taken, with two contact angle values per measurement.

Measurement Name	Value One (°)	Value Two (°)
Unwashed_1	66.18	64.75
Unwashed_2	63.90	63.64
Unwashed_3	63.71	63.81
Ethanol_1	76.58	76.46
Ethanol_2	78.72	78.68
Ethanol_3	76.01	75.85
DMF_1	73.55	73.49
DMF_2	73.42	73.90
DMF_3	73.93	73.83

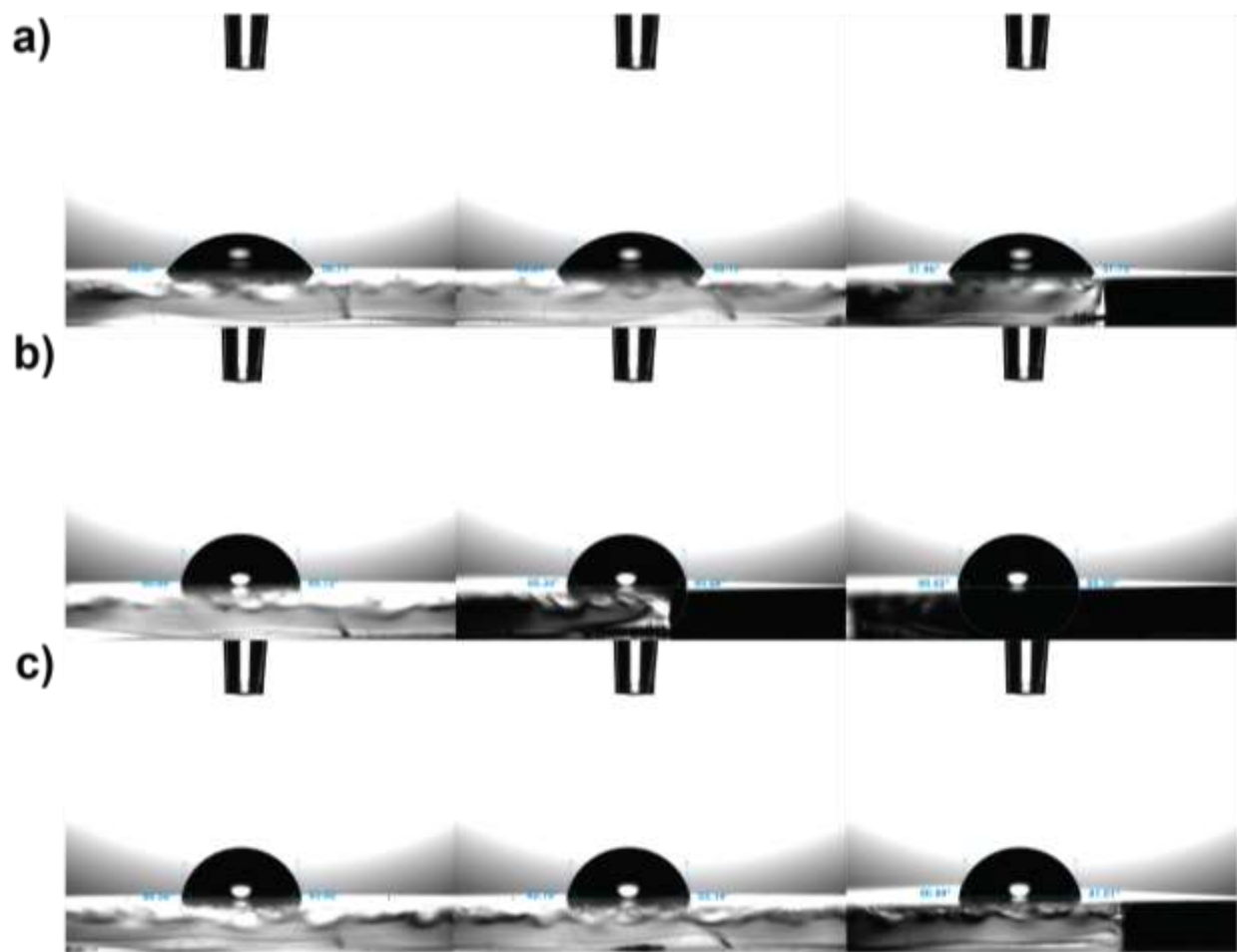


Figure S39: Water contact angle measurements of (a) Unwashed 4PADCB-coated ITO substrates (control), (b) 4PADCB-coated ITO substrates washed with ethanol, and (c) 4PADCB-coated ITO substrates washed with DMF. Each sample had three measurements taken, with each image corresponding to a separate measurement.

Table S4: Water contact angle values on 4PADCB films shown in each image of **Fig. S39**. Each sample had three separate measurements taken, with two contact angle values per measurement.

Measurement Name	Value One (°)	Value Two (°)
Unwashed_1	56.60	56.71
Unwashed_2	56.24	58.13
Unwashed_3	57.96	57.75
Ethanol_1	85.68	85.72
Ethanol_2	85.34	85.08
Ethanol_3	85.52	85.35
DMF_1	84.34	83.82
DMF_2	82.78	83.14
DMF_3	80.89	81.01

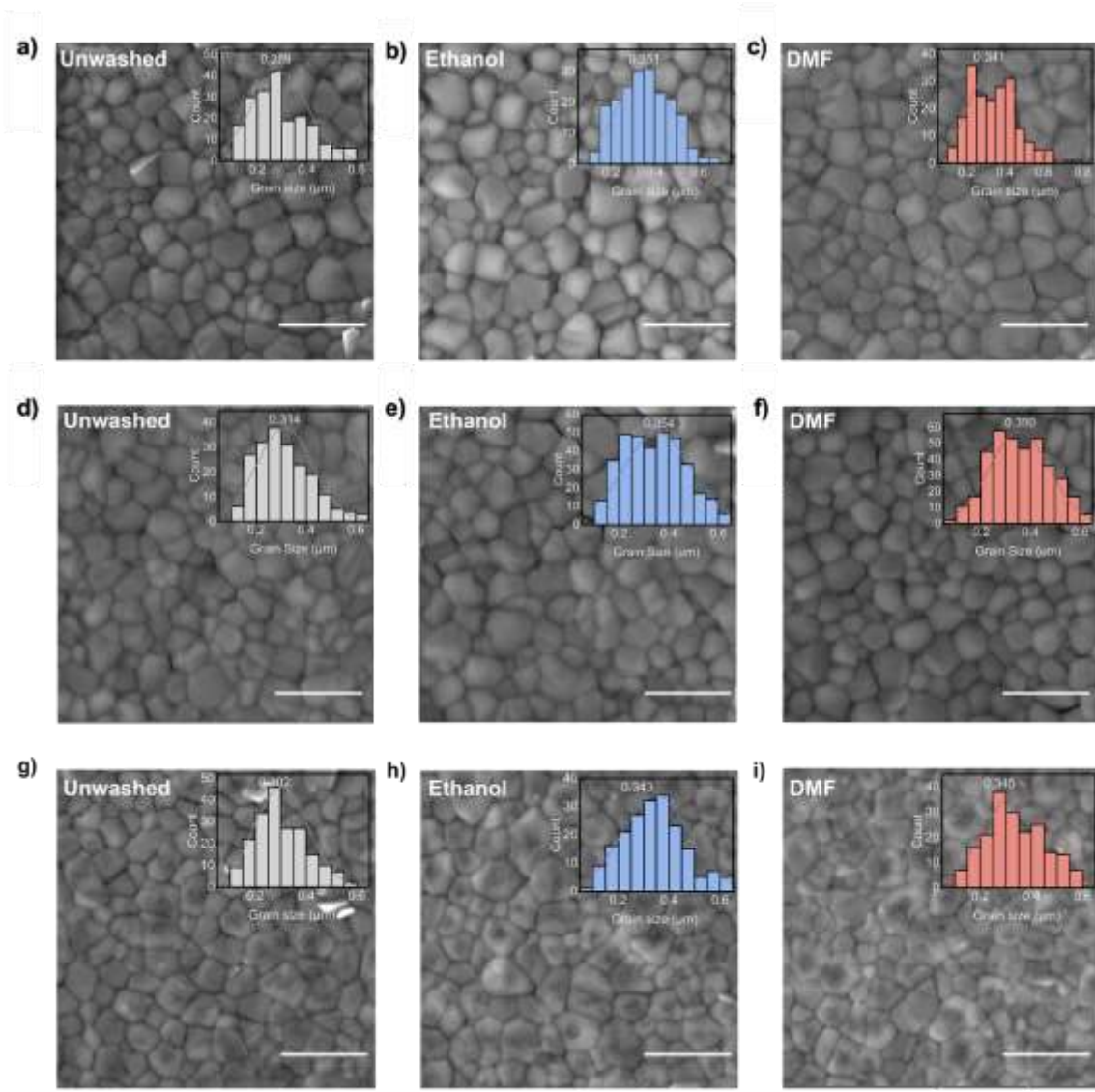


Figure S40: SEM images of perovskite films deposited on (a) unwashed MeO-2PACz, (b) ethanol-washed MeO-2PACz, (c) DMF-washed MeO-2PACz, (d) unwashed 2PACz, (e) ethanol-washed 2PACz, (f) DMF-washed 2PACz, (g) unwashed 4PACB, (h) ethanol-washed 4PACB, and (i) DMF-washed 4PACB. Grain size distributions are shown inset. Scale bars represent 1 μm .

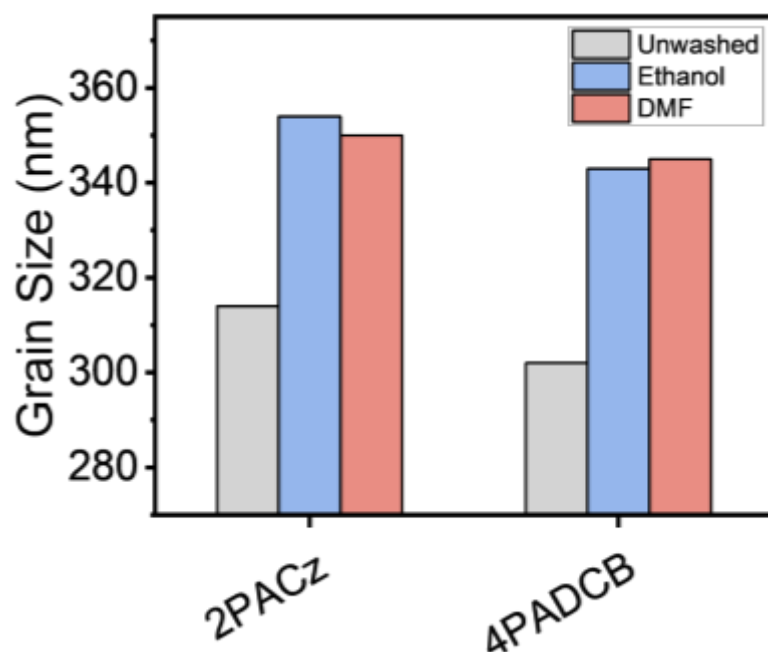


Figure S41: Average grain size of perovskite films deposited on unwashed, ethanol-washed, and DMF-washed (a) 2PACz and (b) 4PADCB.

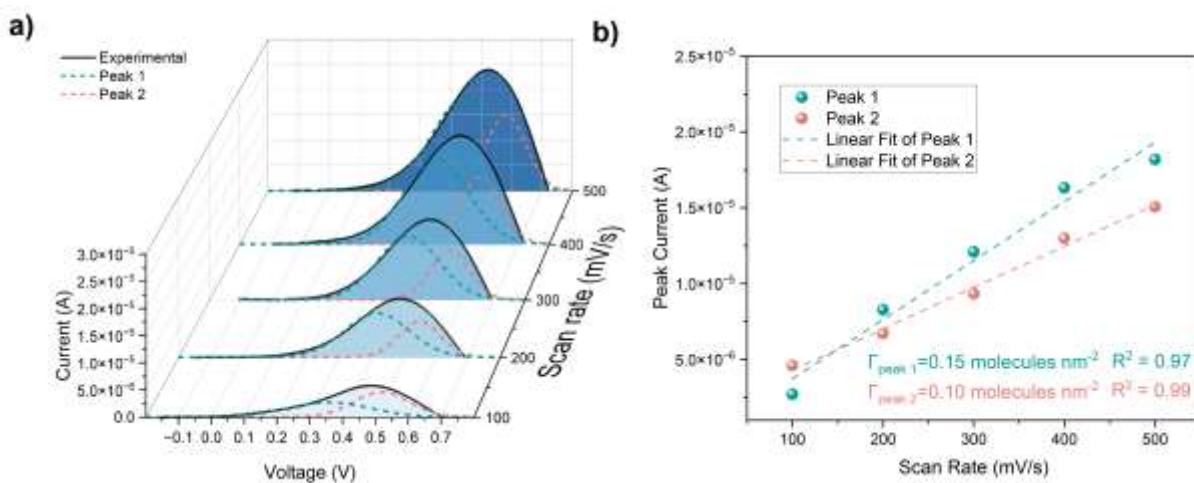


Figure S42: (a) Oxidative peak in cyclic voltammograms of MeO-2PACz-coated ITO substrate after redeposition measured at different voltage scan rates. (b) Correlation between the oxidative peak current and the voltage scan rate for Peak 1 and Peak 2. The dashed lines represent linear fits to the extracted peak currents.

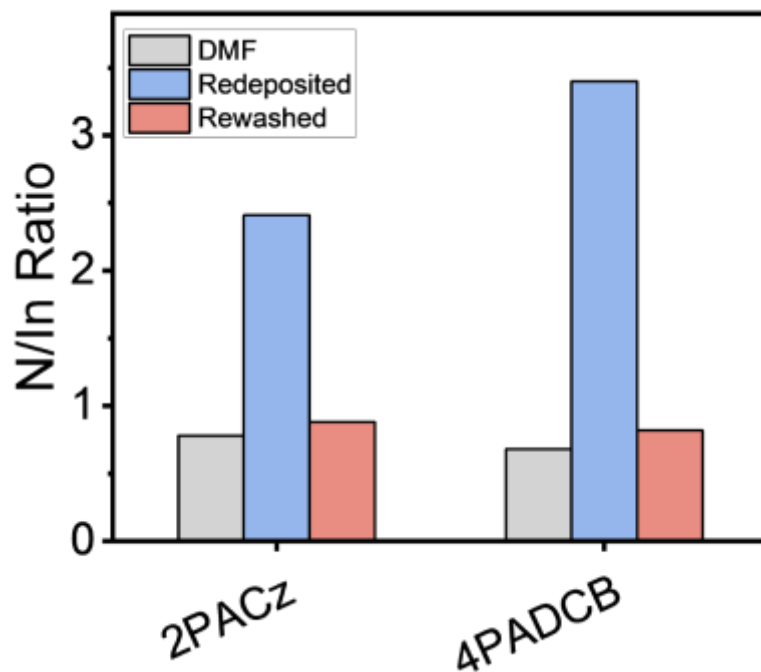


Figure S43: N/In ratio extracted from XPS spectra of DMF-washed, redeposited, and rewashed (a) 2PACz and (b) 4PADCB films.

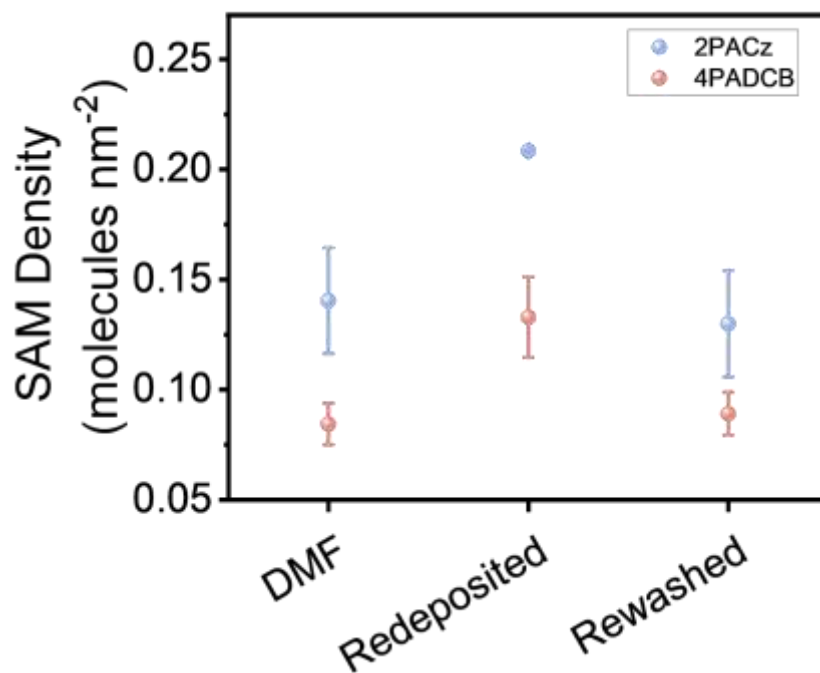


Figure S44: Areal density of DMF-washed, redeposited, and rewashed (a) 2PACz and (b) 4PADCB films extracted from CV.

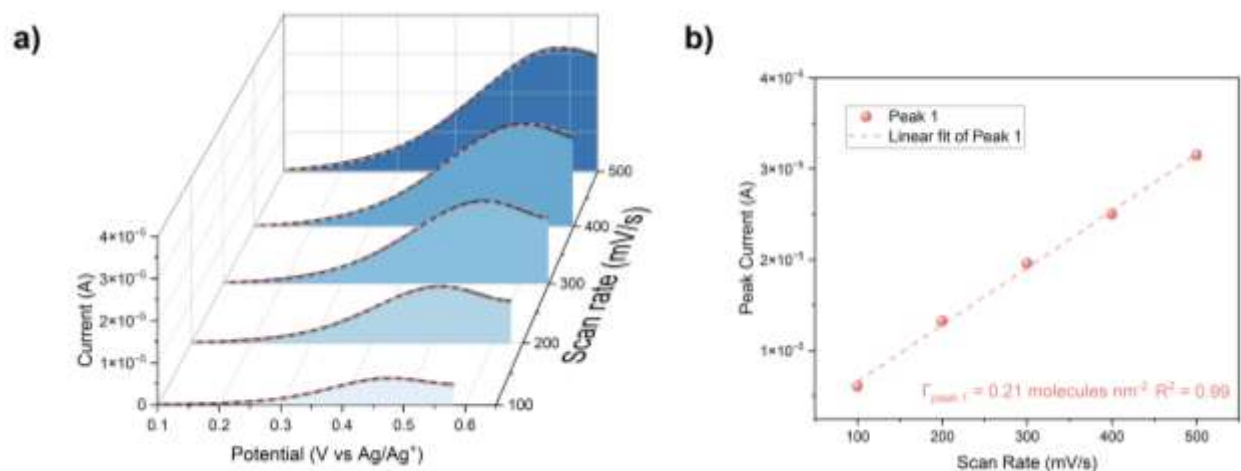


Figure S45: (a) Oxidative peak in cyclic voltammograms of 2PACz-coated ITO substrate after redeposition measured at different voltage scan rates. (b) Correlation between the oxidative peak current and the voltage scan rate. The dashed line represents a linear fit to the extracted peak currents.

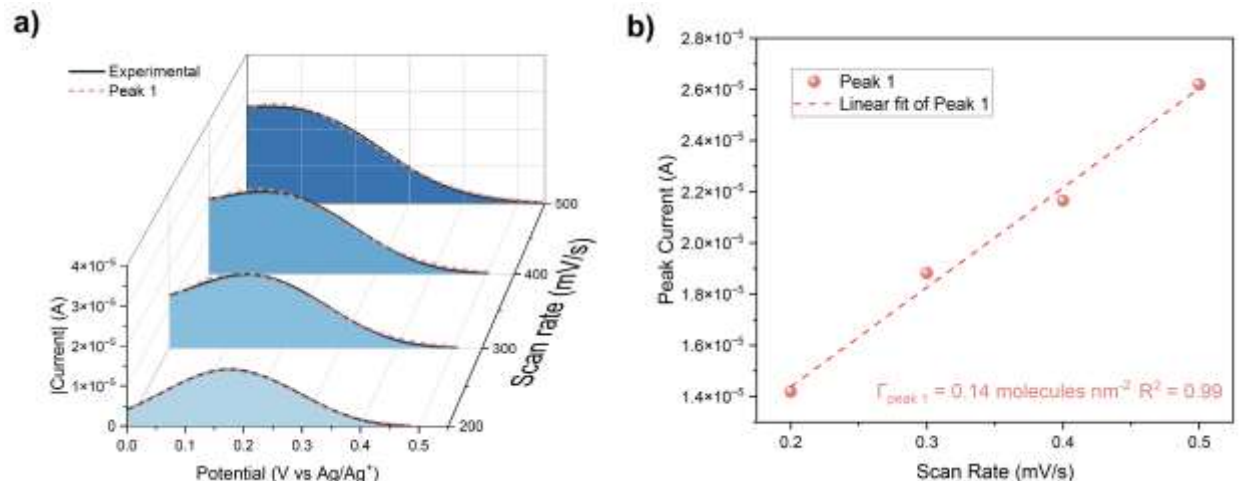


Figure S46: (a) Reductive peak in cyclic voltammograms of 4PADCB-coated ITO substrate after redeposition measured at different voltage scan rates. (b) Correlation between the reductive peak current and the voltage scan rate. The dashed line represents a linear fit to the extracted peak currents.

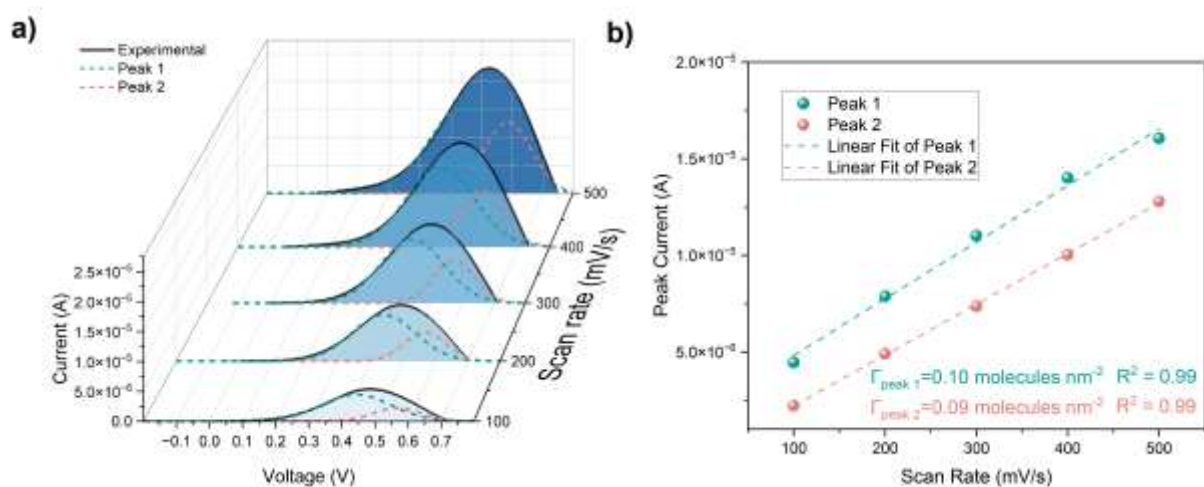


Figure S47: (a) Oxidative peak in cyclic voltammograms of MeO-2PACz-coated ITO substrate after rewashing measured at different voltage scan rates. (b) Correlation between the oxidative peak current and the voltage scan rate for Peak 1 and Peak 2. The dashed lines represent linear fits to the extracted peak currents.

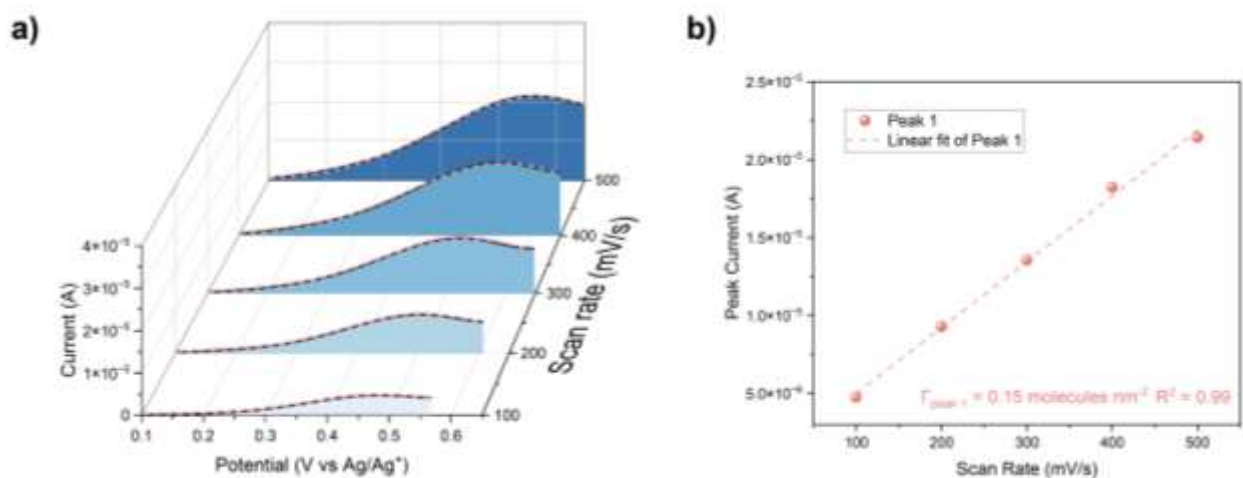


Figure S48: (a) Oxidative peak in cyclic voltammograms of 2PACz-coated ITO substrate after rewashing measured at different voltage scan rates. (b) Correlation between the oxidative peak current and the voltage scan rate. The dashed line represents a linear fit to the extracted peak currents.

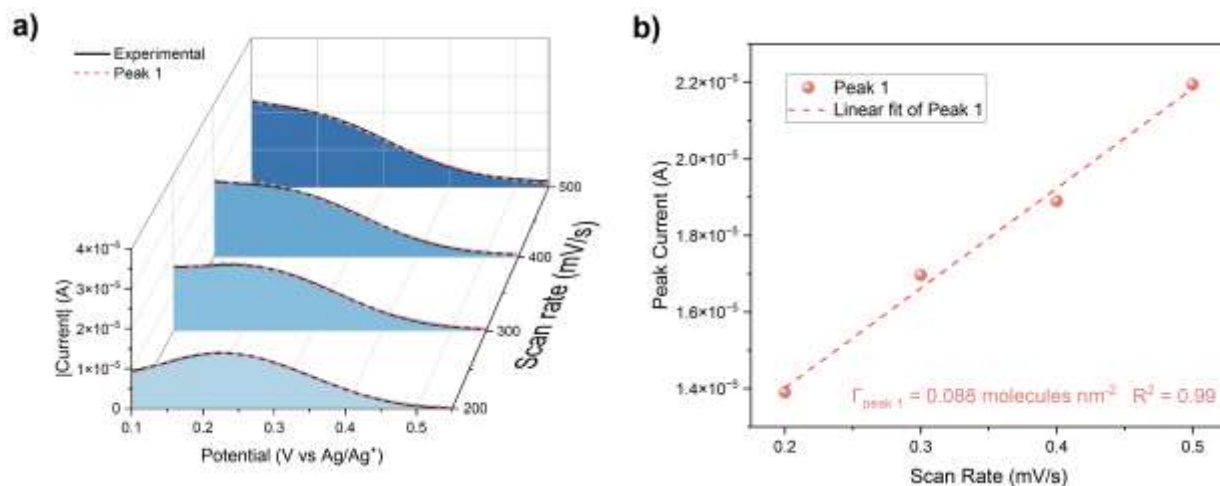


Figure S49: (a) Reductive peak in cyclic voltammograms of 4PADCB-coated ITO substrate after rewashing measured at different voltage scan rates. (b) Correlation between the reductive peak current and the voltage scan rate. The dashed line represents a linear fit to the extracted peak currents.

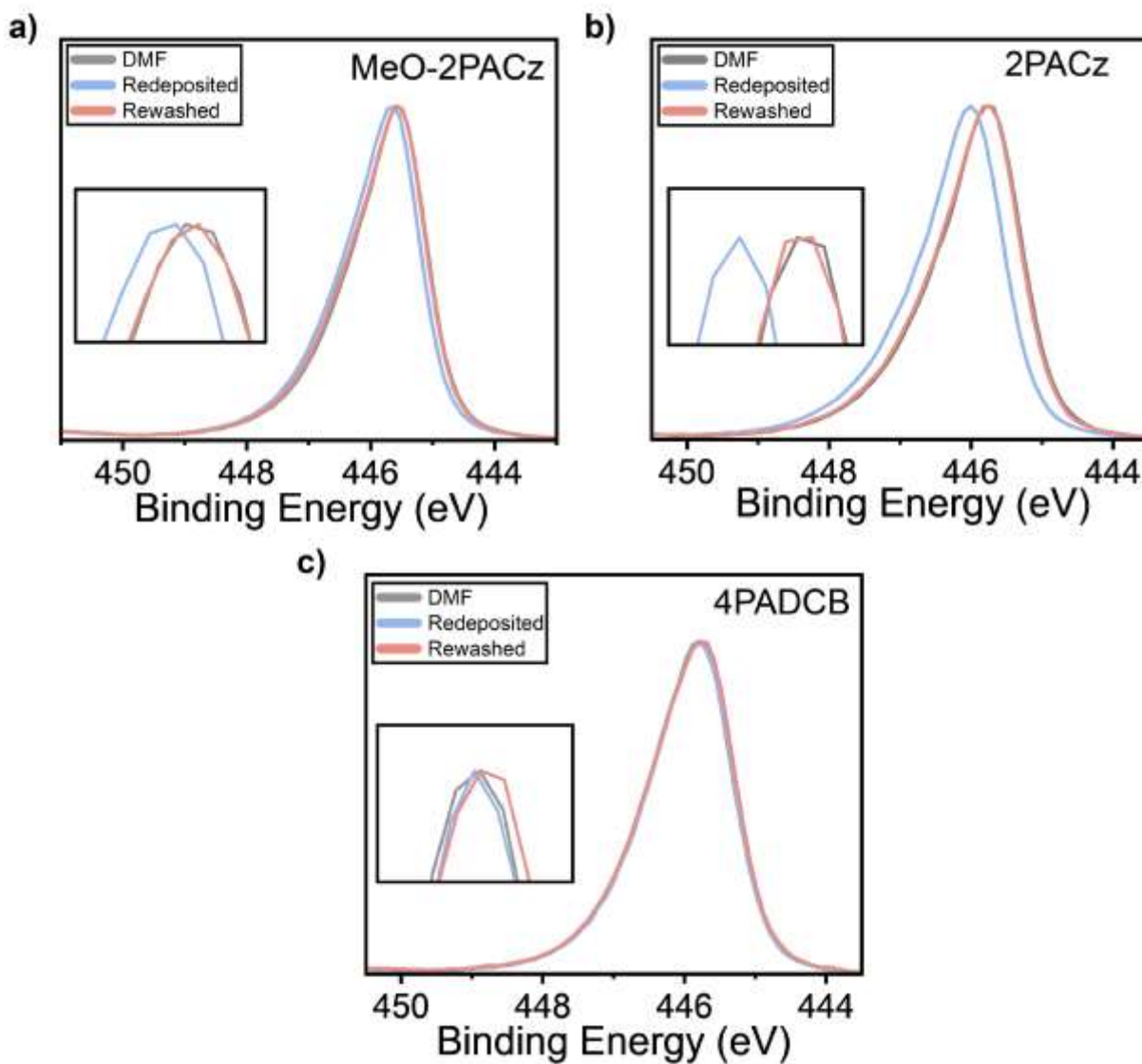


Figure S50: In 3d_{5/2} XPS spectra for DMF-washed, redeposited, and rewashed (a) MeO-2PACz, (b) 2PACz, and (c) 4PADCB. Peak shifts are highlighted inset.

Supplementary Note 2: To gain further insight into how redeposited SAM molecules interact with the ITO surface, we examined shifts in the In 3d_{5/2} core level using XPS (**Fig. S50**). Compared to DMF-washed MeO-2PACz sample, redeposition induced a blue shift of +0.08 eV, consistent with renewed adsorption of SAMs. After rewashing, however, the peak returned to nearly the same binding energy as the DMF-washed sample, indicating that the molecules causing the blue shifting were weakly adsorbed. While this same behavior was observed for 2PACz, for 4PADCB, we observed negligible peak shifting after redeposition and rewashing, supporting the conclusion that, after the initial DMF wash, 4PADCB molecules adopt a flatter orientation that prevents subsequent redeposited molecules from effectively accessing the substrate. Nonetheless, these findings further demonstrate that the redeposited SAM molecules resistant to solvent-induced removal do not form covalent bonds with the ITO substrate.

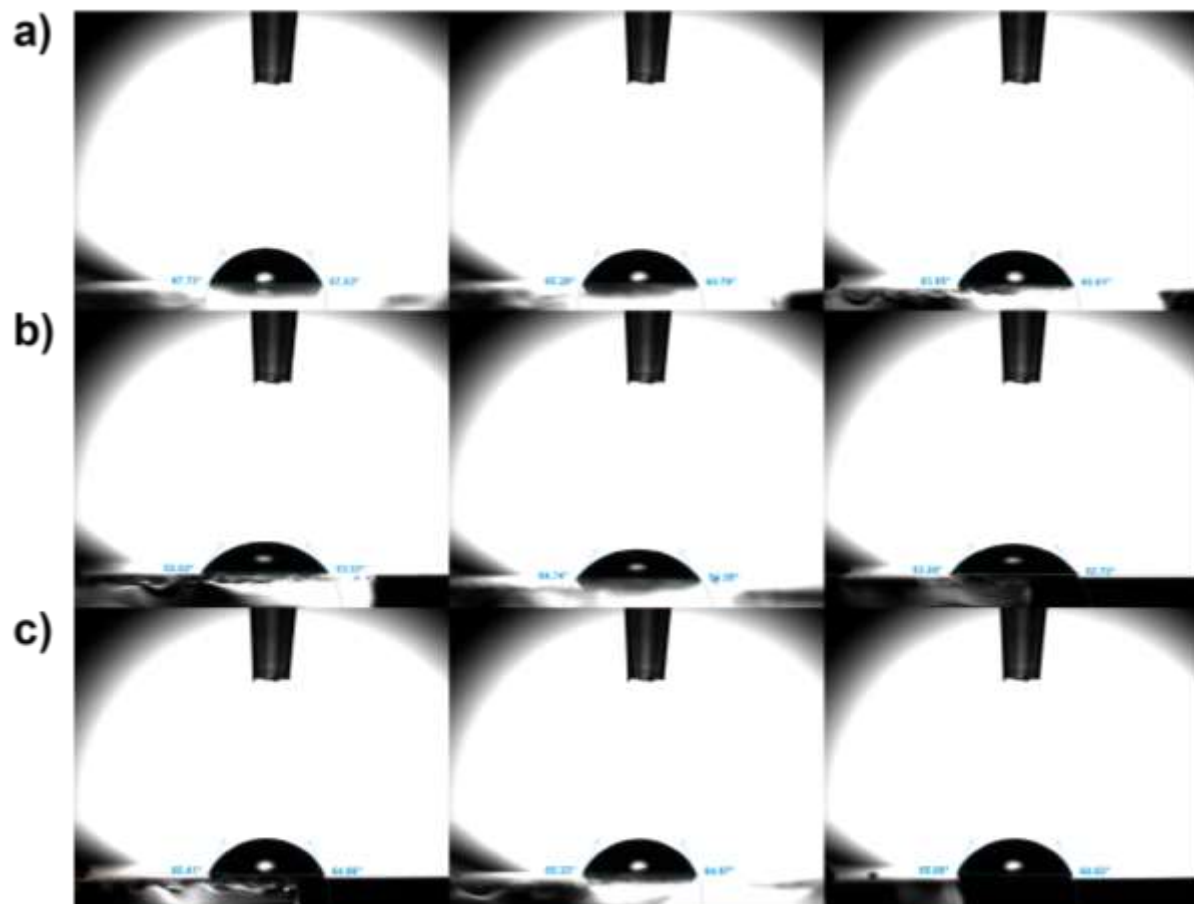


Figure S51: Water contact angle images of MeO-2PACz films that are (a) DMF-washed, (b) redeposited, and (c) rewashed. Each sample had three measurements taken, with each image corresponding to a separate measurement.

Table S5: Water contact angle values on MeO-2PACz films shown in each image of **Fig. S51**. Each sample had three separate measurements taken, with two contact angle values per measurement.

Measurement Name	Value One (°)	Value Two (°)
DMF_1	67.73	67.63
DMF_2	65.29	64.79
DMF_3	63.85	63.61
Redeposited_1	53.82	53.57
Redeposited_2	54.74	54.35
Redeposited_3	53.09	52.72
Rewashed_1	65.41	64.86
Rewashed_3	65.33	64.97
Rewashed_3	65.08	64.83

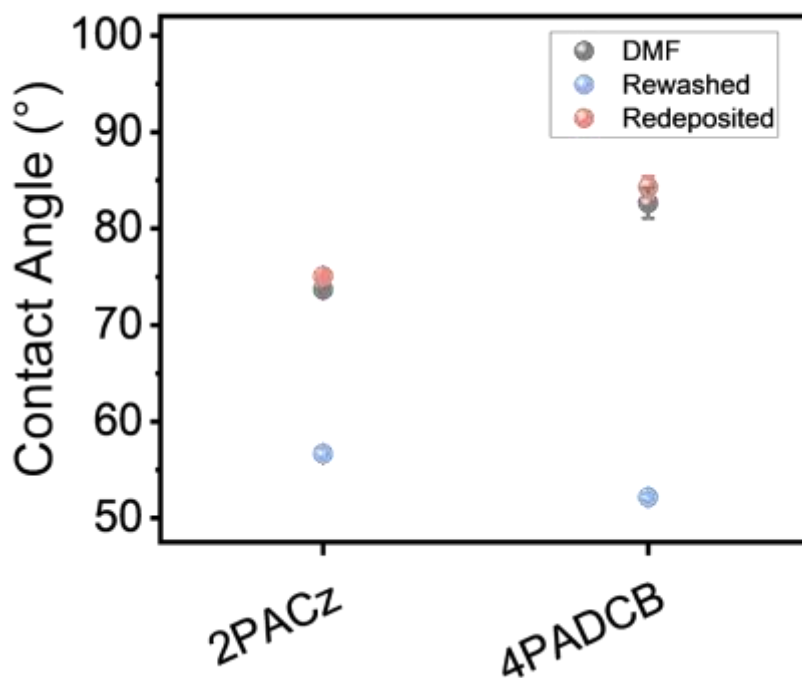


Figure S52: Average water contact angle of DMF-washed, redeposited, and rewashed 2PACz and 4PADCB.

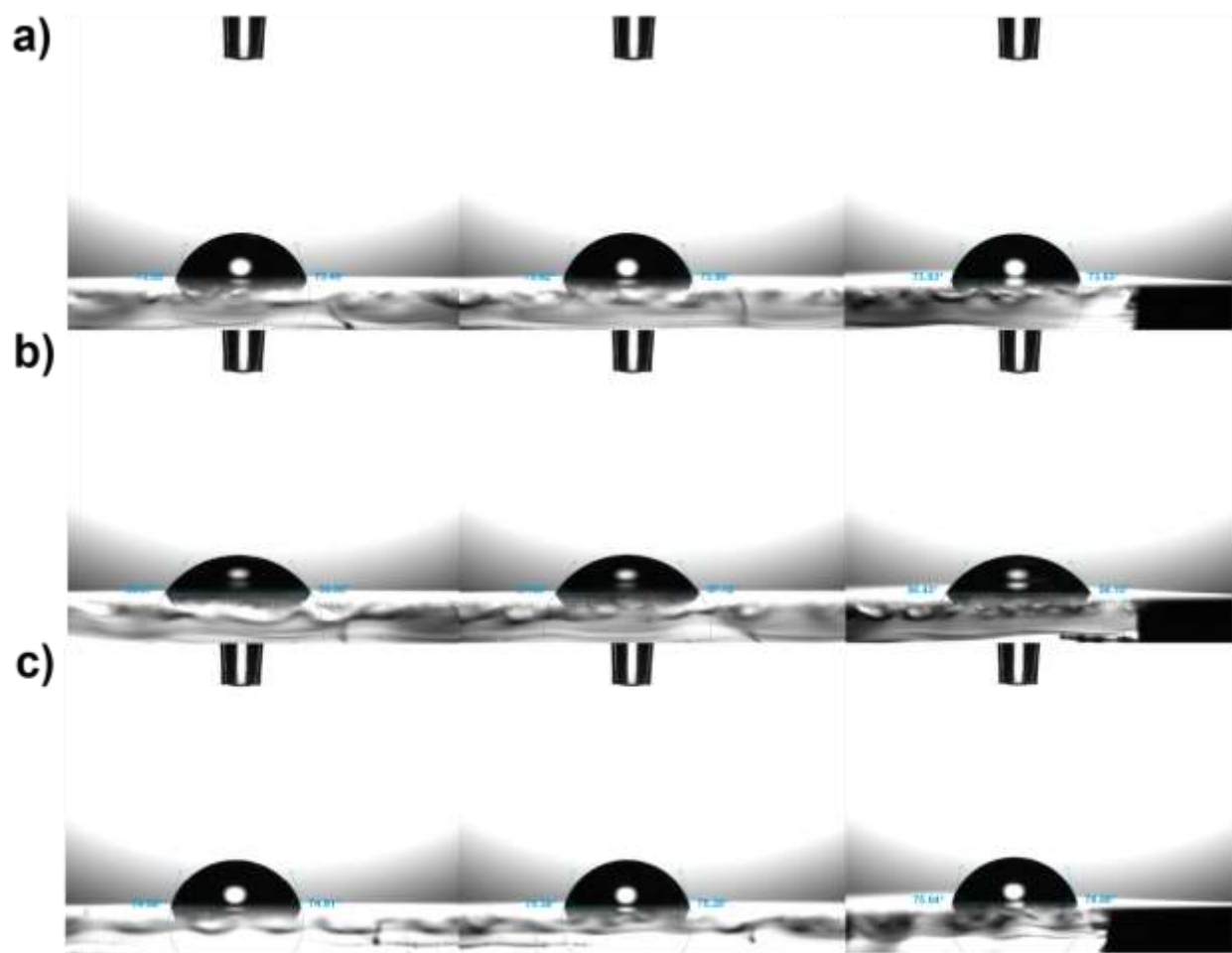


Figure S53: Water contact angle images of 2PACz films that are (a) DMF-washed, (b) redeposited, and (c) rewashed. Each sample had three measurements taken, with each image corresponding to a separate measurement.

Table S6: Water contact angle values on 2PACz films shown in each image of **Fig. S53**. Each sample had three separate measurements taken, with two contact angle values per measurement.

Measurement Name	Value One (°)	Value Two (°)
DMF_1	73.55	73.49
DMF_2	73.42	73.90
DMF_3	73.93	73.83
Redeposited_1	56.07	56.86
Redeposited_2	57.48	57.12
Redeposited_3	56.43	56.10
Rewashed_1	74.98	74.91
Rewashed_3	75.38	75.28
Rewashed_3	75.04	74.88

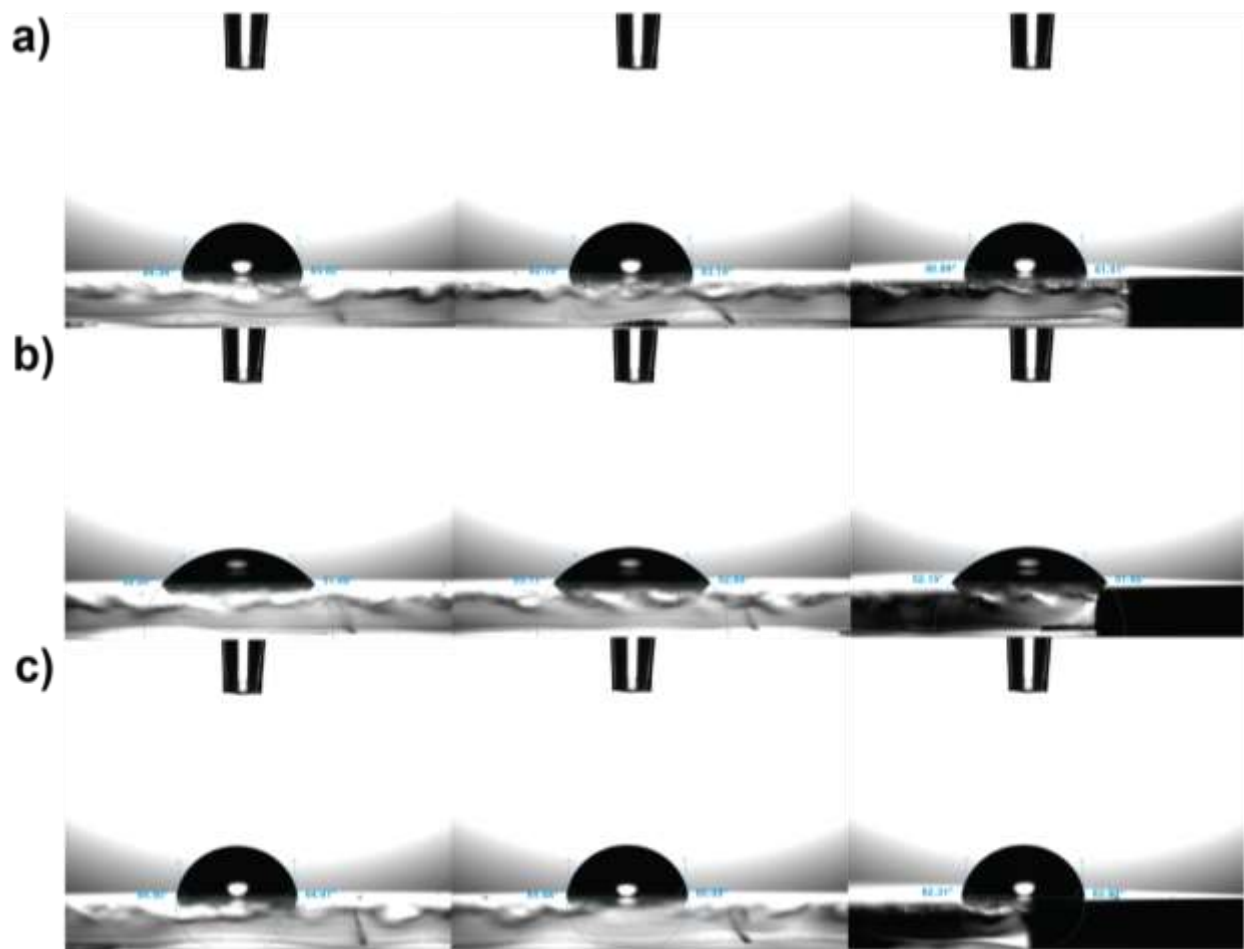


Figure S54: Water contact angle images of 4PADCB films that are (a) DMF-washed, (b) redeposited, and (c) rewashed. Each sample had three measurements taken, with each image corresponding to a separate measurement.

Table S7: Water contact angle values on 4PADCB films shown in each image of **Fig. S54**. Each sample had three separate measurements taken, with two contact angle values per measurement.

Measurement Name	Value One (°)	Value Two (°)
DMF_1	84.34	83.82
DMF_2	82.78	83.14
DMF_3	80.89	81.01
Redeposited_1	52.20	51.49
Redeposited_2	53.11	52.05
Redeposited_3	52.15	51.92
Rewashed_1	84.50	84.67
Rewashed_2	85.44	85.09
Rewashed_3	82.31	83.90

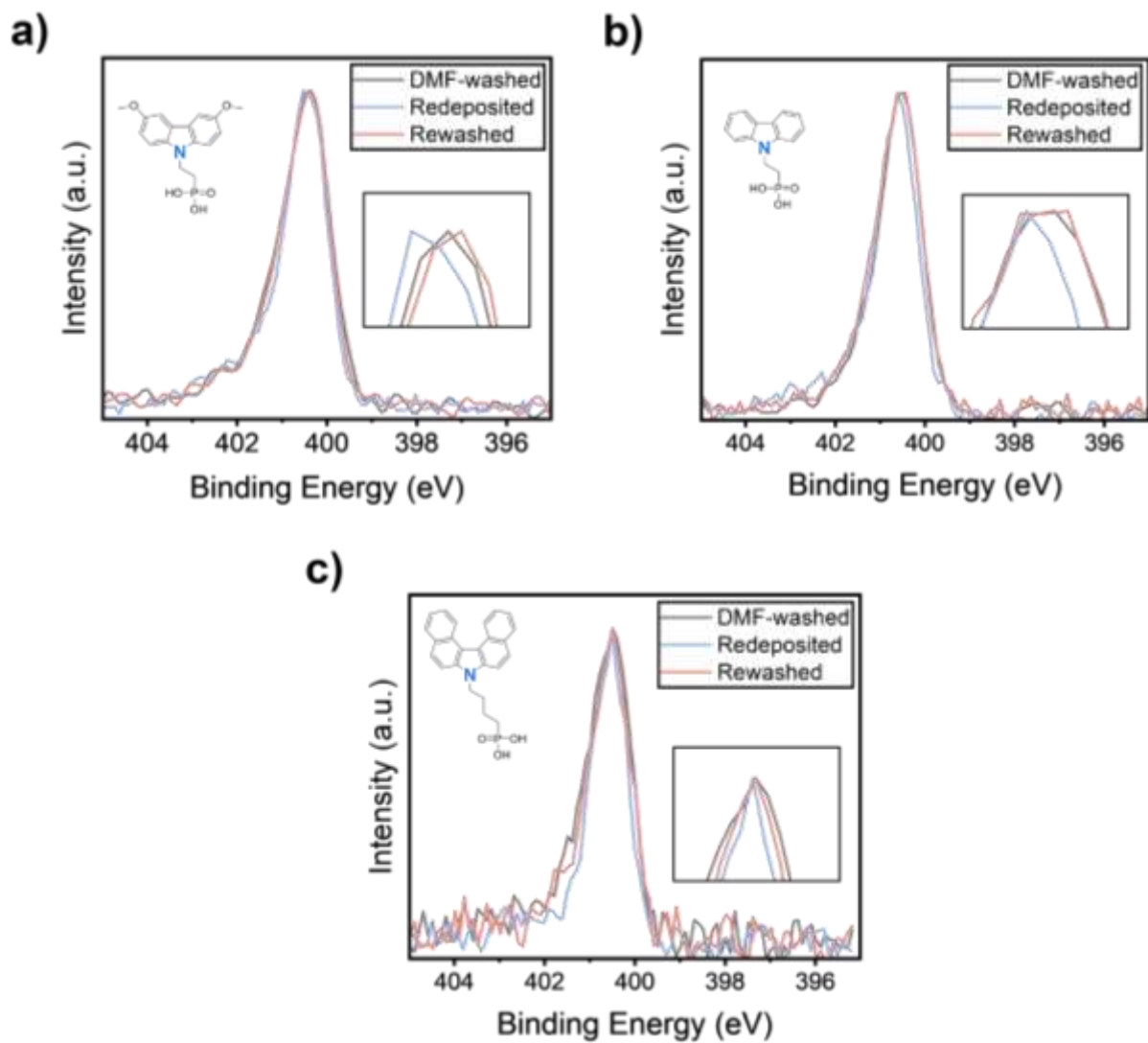


Figure S55: N 1s XPS spectra of DMF-washed, redeposited, and rewashed (a) MeO-2PACz, (b) 2PACz, and (c) 4PADCB samples, with peak shifts highlighted in the inset.

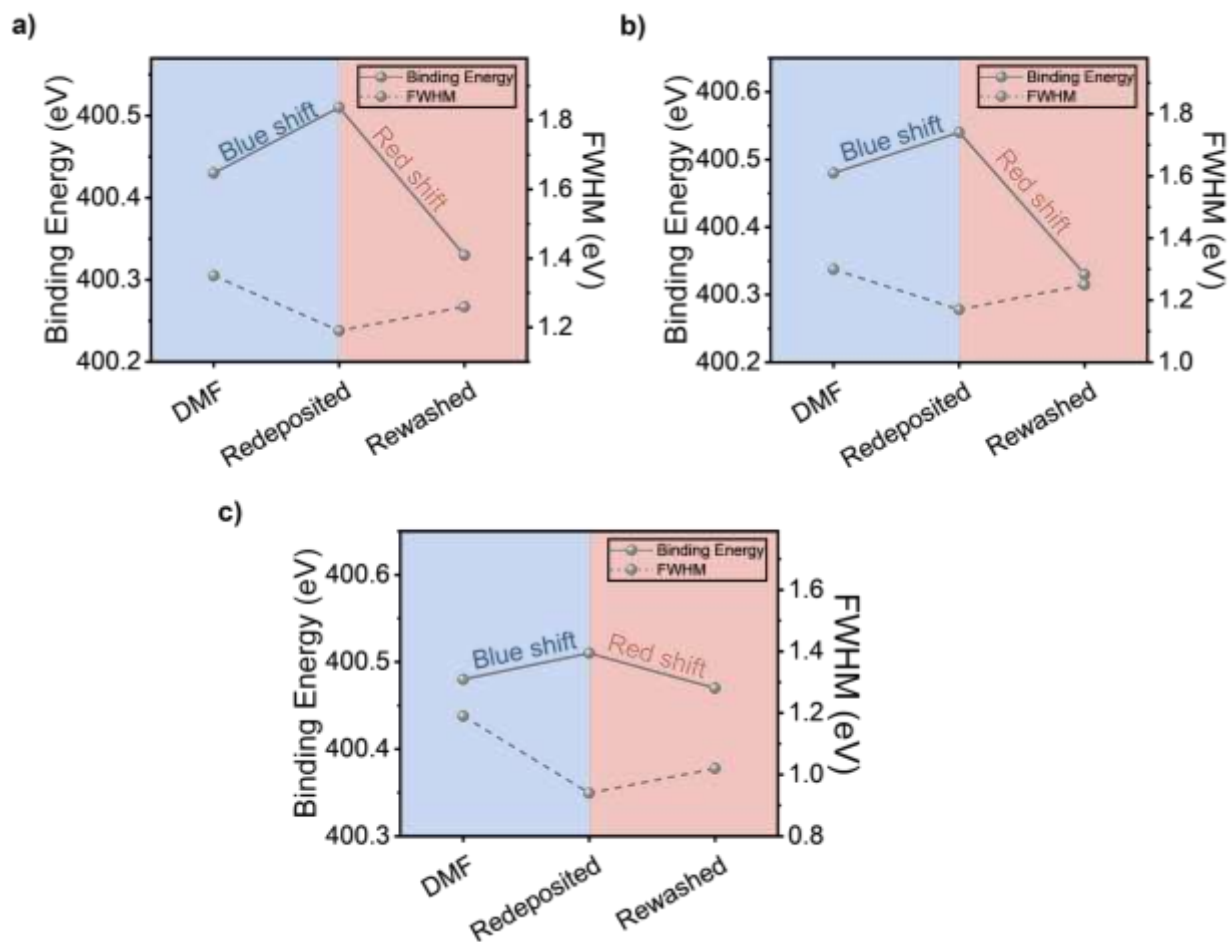


Figure S56: Binding energy and full width at half maximum extracted from N 1s XPS spectra of DMF-washed, redeposited, and rewashed (a) MeO-2PACz, (b) 2PACz, and (c) 4PADCB films.

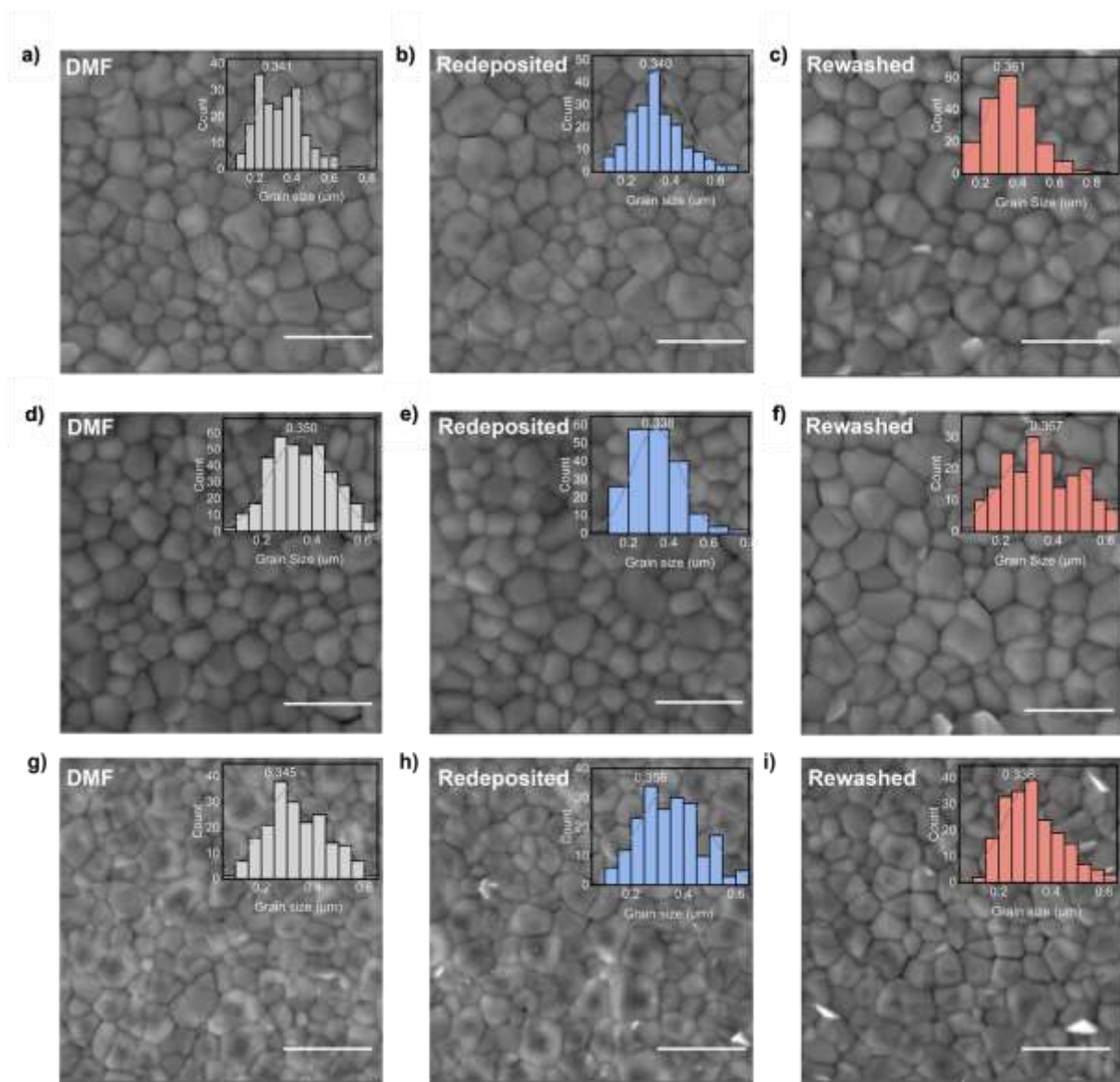


Figure S57: SEM images of perovskite films deposited on (a) DMF-washed MeO-2PACz, (b) redeposited MeO-2PACz, (c) rewashed MeO-2PACz, (d) DMF-washed 2PACz, (e) redeposited 2PACz, and (f) rewashed 2PACz, (g) DMF-washed 4PADCB, (h) redeposited 4PADCB, and (i) rewashed 4PADCB. Grain size distributions are shown inset. Scale bars represent 1 μm .

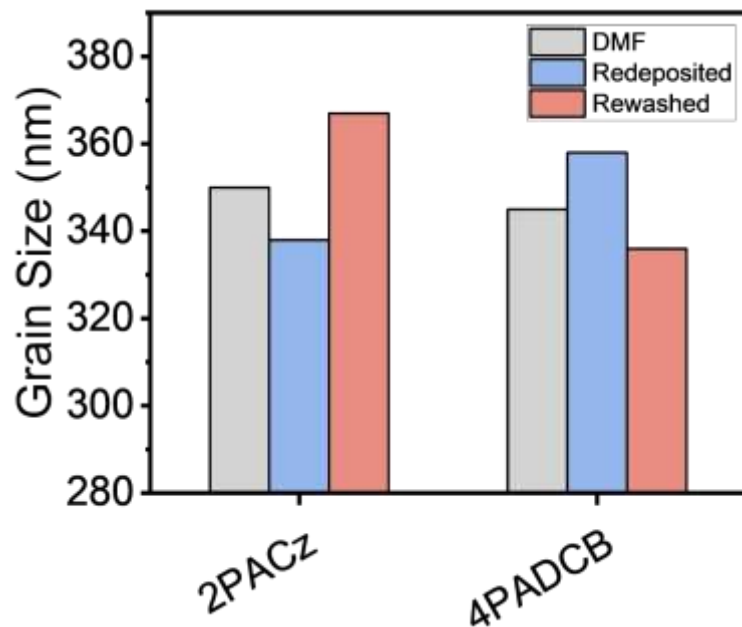


Figure S58: Average grain size of perovskite films deposited on DMF-washed, redeposited, and rewashed 2PACz and 4PADCB.

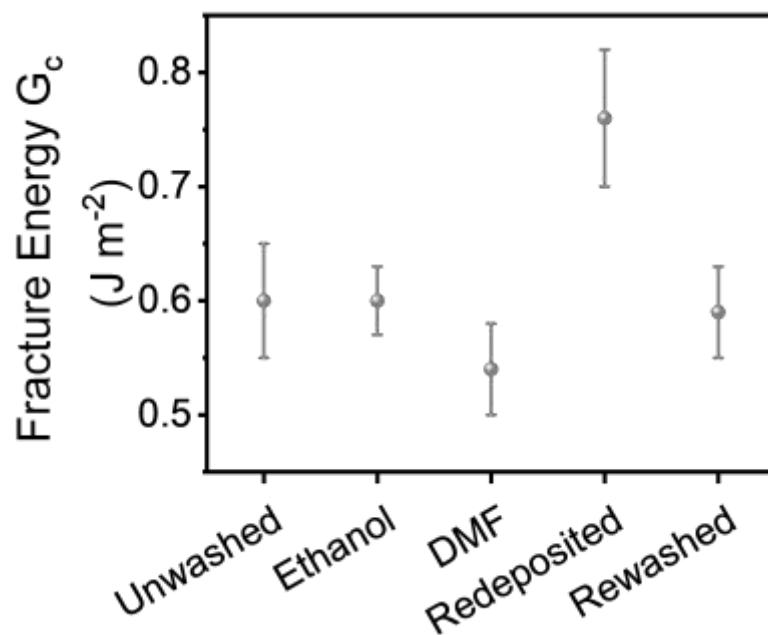


Figure S59: Fracture energy (G_c) values of the MeO-2PACz/perovskite interface in unwashed, ethanol-washed, DMF-washed, redeposited, and rewashed samples.

Supplementary Note 3: We conducted G_c measurements to examine how SAM–perovskite interactions evolve under different washing and redeposition conditions. As shown in **Fig. S59**, washing led to slight reductions in G_c , consistent with the removal of SAM molecules and the resulting decrease in available SAM–perovskite interaction sites. Following redeposition, G_c increased, suggesting that additional SAM molecules were present at the surface. However, a subsequent rewash reduced G_c back to a value comparable to that of the unwashed sample. This trend indicates that SAM abundance alone does not explain the G_c increase after redeposition; if it did, the unwashed sample would exhibit a higher G_c than the rewash sample. We therefore attribute the observed increase to a combination of enhanced SAM abundance and improved molecular ordering, including a higher proportion of molecules with terminal functional groups favorably interacting with, and oriented toward, the perovskite surface, which collectively strengthen SAM–perovskite interfacial interactions.

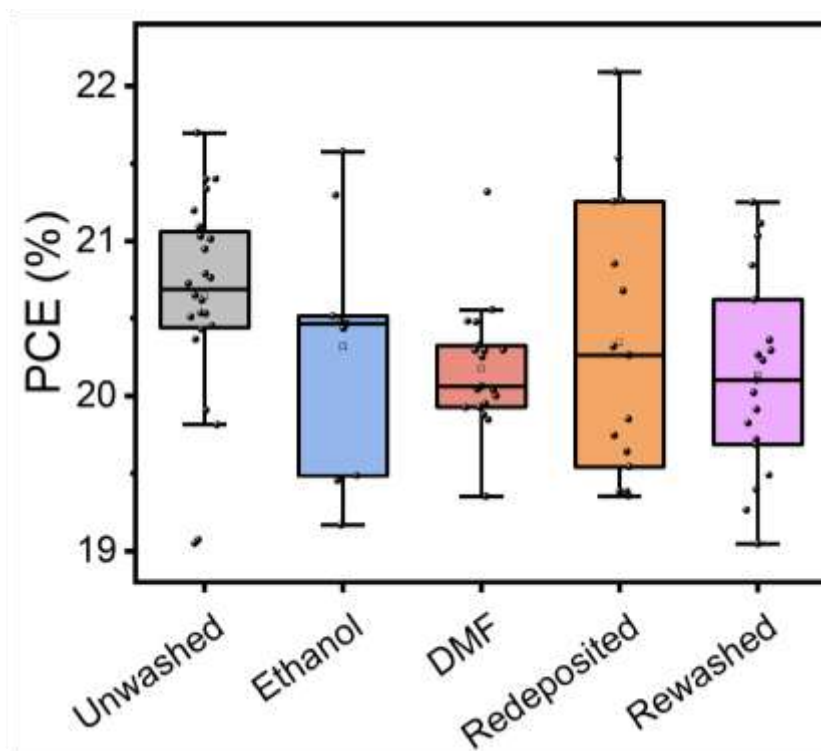


Figure S60: Performance parameters of perovskite solar cells with MeO-2PACz as the SAM HTL.

Table S8: Photovoltaic parameters of perovskite solar cells employing MeO-2PACz as the SAM-based HTL. The data includes device yield (fraction of working devices over total), maximum PCE, average PCE with standard deviation, and the corresponding average J_{sc} , V_{oc} , FF, and hysteresis (i.e., difference between forward scan and reverse scan PCE values), each reported with standard deviation.

	Unwashed	Ethanol	DMF	Redeposited	Rewashed
Device Yield	24/40 (60%)	9/24 (38%)	19/32 (59%)	15/24 (63%)	19/32 (59%)
Maximum PCE (%)	21.7	21.6	21.3	22.1	21.3
Average PCE (%)	20.6 ± 0.7	20.3 ± 0.8	20.2 ± 0.4	20.3 ± 0.9	20.1 ± 0.6
J_{sc} (mA cm⁻²)	20.9 ± 0.6	21.1 ± 0.5	20.5 ± 0.3	21.1 ± 0.8	20.7 ± 0.5
V_{oc} (V)	1.01 ± 0.01	1.01 ± 0.02	1.02 ± 0.01	1.01 ± 0.01	1.00 ± 0.01
FF (%)	79.2 ± 1.9	77.2 ± 3.0	78.7 ± 1.9	77.6 ± 1.5	78.7 ± 1.2
Hysteresis (%)	0.3 ± 0.4	0.2 ± 0.1	0.2 ± 0.1	0.5 ± 0.2	0.5 ± 0.2

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